

Table 1001 to 1500 (continued)

The following pages reproduce the substance of the printed table 1001–1500 of the Pellian Equation report. For each entry the continued-fraction expansion of  $\sqrt{a}$  is given on the upper line of the cell, the period being indicated by enclosing the central term (or pair of terms, marked with \*) in parentheses; the lower line gives the second-row auxiliaries. The right-hand column gives the corresponding integer solution  $(y, x)$  of  $y^2 - ax^2 = \pm 1$ . Stars on the item-number mark entries where the period has a double middle term.

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines                                                          | $y \quad / \quad x$   |
|-------|-----|-----------------------------------------------------------------------------------|-----------------------|
| 1147  | 33; | 8, 9, 34, 33, 1, 28, 27, 33, 26, 41, 3, (62)                                      | 2789 45403            |
|       | 58, | 9, 34, 33, 1, 28, 27, 33, 26, 41, 3, ...                                          | 94471 524381986       |
| 1148  | 33; | 1, 7, 2, (16)                                                                     | [948]                 |
|       | 59, | 8, 31, (4)                                                                        |                       |
| 1149  | 33; | 1, 8, 1, 2, 3, (22)                                                               | 1 65887               |
|       | 60, | 7, 44, 21, 20, (3)                                                                | 2 017226921547        |
| 1150  | 33; | 1, 10, 3, 7, 4, 1, 2, (2)                                                         | 349 25592             |
|       | 61, | 6, 21, 9, 14, 41, 21, (25)                                                        | 11843 84449           |
| 1151  | 33; | 1, 12, 1, 1, 2, 2, 3, 6, 2, (33)                                                  | 19426 07107           |
|       | 62, | 5, 38, 29, 23, 25, 19, 10, 31, (2)                                                | 6 59056 71854707      |
| 1152  | 33; | 1, (15)                                                                           | 17                    |
|       | 63, | (4)                                                                               | 577                   |
| 1153* | 33; | 1, 21, 1, 1, 1, 6, 1, 7, 1, 1, 1, 1, 1, 2, 4, 1, 5, 2, 1, 3, (1, 1)               | 3 01789 02568 75073   |
|       | 64, | 3, 43, 24, 41, 9, 53, 8, 39, 27, 32, 29, 33, 24, 13, 48, 11, 23, 39, 16, (33, 33) | 102 47504 00230 72656 |
| 1154  | 33; | 1, (32)                                                                           | 1155                  |
|       | 65, | (2)                                                                               | 341                   |
| 1155  | 33; | (1)                                                                               | 34                    |
|       |     | (66)                                                                              | 1                     |
| 1157* | 34; | (6, (81))                                                                         | 115374                |
|       |     |                                                                                   | 17                    |
| 1158  | 34; | (3, (41))                                                                         | 4902 89385 75180      |
|       |     |                                                                                   | 1 66914 55514 24551   |
| 1159  | 34; | 22, 1, 2, 7, 4, 2, 2, 13, 4, 1, 3, 1, (2)                                         | 579                   |
| 1160  | —   |                                                                                   | —                     |
| 1161  | 3,  | 45, 22, 9, 15, 25, 27, 5, 14, 45, 15, 42, (19)                                    | 56 21214 40972        |
|       |     |                                                                                   | 1915 34168 54935      |
| 1162a | 34; | (17) (4)                                                                          | 2266 6136416723       |
| 1162b | 34; | 13, 1, 1, 1, 1, 2, 8, 7, 2, 4, 1, 3, (2)                                          | 369 56541             |
|       | 5,  | 40, 27, 31, 32, 25, 8, 9, 29, 13, 45, 16, (27)                                    | 12603 214090320       |
| 1162c | 34; | 11, 2, 1, 6, 1, (8)                                                               | 1 68199               |
|       | 6,  | 23, 42, 9, 54, (7)                                                                | 867 20773             |
| 1163  | 34; | 9, 1, 2, 1, 2, 4, 1, 1, (33)                                                      | 29599 61778           |
| 1164  | 7,  | 46, 19, 38, 23, 14, 31, 37, (2)                                                   | 193 40870             |
| 1164b | 34; | 8, 1, 1, (16)                                                                     | 6604 27701            |
|       | 8,  | 33, 35, (4)                                                                       | 10943                 |
| 1165* | 34; | 7, 1, 1, 3, 16, 1, 3, 1, (1, 1)                                                   | 3 73828               |
|       | 9,  | 36, 31, 19, 4, 51, 15, 36, (29, 29)                                               | 22 6861725409         |
| 1166  | 34; | 6, 1, 4, 2, 1, 1, 9, (6)                                                          | —                     |
|       | 10, | 49, 13, 25, 29, 38, 7, (11)                                                       | —                     |
| 1167  | 34; | 6, 5, (11)                                                                        | —                     |
|       | 11, | 13, (6)                                                                           | —                     |
| 1168  | 34; | 5, 1, 2, 7, (4)                                                                   | —                     |
|       | 12, | 41, 23, 9, (16)                                                                   | —                     |

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines                                                                | $y / x$                       |
|-------|-----|-----------------------------------------------------------------------------------------|-------------------------------|
| 1169  | 34; | 5, 4, 13, 2, 3, 1, 1, 5, 1, 1, 1, 7, 1, (8)                                             | 538 84918 28064               |
|       |     | 13, 16, 5, 20, 17, 32, 35, 11, 40, 25, 41, 8, 55, (7)                                   | 18423 59949 83935             |
| 1170  | 34; | 1, 29, 17, 32, 35, 1, 40, 25, 41, 8, 55, (7)                                            | 6499                          |
| 1171  | 34; | 5, 4, ((69))                                                                            | —                             |
| 1172  | —   |                                                                                         | —                             |
| 1173  | 34; | 4, 49, 51, ...                                                                          | 15274 30263                   |
|       |     | 15, —                                                                                   | 5 22684                       |
| 1174  | 34; | —                                                                                       | 1233 5769183900               |
|       |     | 13, 42, 21, 26, 17, 35, 30, 27, 41, (2)                                                 | 20614397                      |
| 1175  | 34; | 16, 33, 34, 3, 1, 1, (16)                                                               | —                             |
|       |     | 16, 31, 37, (4)                                                                         | —                             |
| 1176  | 34; | (4, 4) 1, 43, —                                                                         | 4                             |
| 1177  | 34; | 18, 15, 14, 1, 4, 2, 13, 3, 1, 22, 11, 2, 1, 1, 1, 6, 4, 2, 2, 1, 1, 7, (34)            | 1363 67209 12136 61406 74774  |
| 1178  |     | 3, 43, 1, 2, 6, 13, 30, 5, 17, 50, 3, 6, 25, 34, 27, 39, 10, 15, 26, 23, 30, 37, 9, (2) | 46724 42879 20656 93035 91365 |
| 1179  | 34; | 19, 3, 2, 5, (1)                                                                        | 12901                         |
|       |     | 34, 25, 11, (50)                                                                        | 4 42212440                    |
| 1180  | 34; | 20, 25, 31, (24)                                                                        | 4801                          |
| 1181* | 34; | 21, 16, (2) 22, 9, 1, 3, 7, 2, 1, 2, 1, 1, 2, 2, 2, 8, (6)                              | 276 38354 60657 78460         |
| 1182  | —   |                                                                                         | 9482 01015 39044              |
| 1183  |     | 3, 31, 3, 7, 48, 17, 9, 24, 37, 21, 33, 32, 23, 24, 27, 8, (11)                         | —                             |
| 1184  | —   |                                                                                         | 751375316                     |
| 1185  | 34; | 22, (2)                                                                                 | 309                           |
| 1186  | 34; | 23, 9, (33)                                                                             | 10610                         |
|       |     | 24, 45, —                                                                               | 20 24999                      |
| 1187  | —   |                                                                                         | 58950                         |
| 1188  | —   |                                                                                         | —                             |
| 1189* | —   |                                                                                         | —                             |
|       |     | 1, 20, 1, 36, 31, 21, 39, (20)                                                          | —                             |
| 1189b | 34; | 25, 2, 1, 3, 3, 5, 1, 16, 2, 1, (13, 13)                                                | 35 62788 38045                |
|       |     | 37, 29, 1, 41, 17, 20, 11, 55, 4, 23, 44, (5, 5)                                        | 1224 37647 36718              |
| 1189c | 34; | 26, 29, 1, 2, 2, 1, 2, 1, (10)                                                          | 2727 5922568872               |
|       |     | 34, 23, 22, 39, 19, 47, (6)                                                             | 37                            |
| 1189d | 34; | 27, 33, 2, 7, 22, 1, (3)                                                                | 1305                          |
| 1189e |     | 38, 1, 9, (17)                                                                          | 96401                         |
|       |     |                                                                                         | 76328                         |
| 1189f | 34; | 28, 25, 2, 16, 49, 7, (4)                                                               | 9655220815                    |
|       |     | 9, 3, 53, (14)                                                                          | 41142 0138725010              |
| 1189g | 34; | 2, 21, 1, 3, 1, 1, 2, 3, (4)                                                            | —                             |
|       |     | 29, 3, 16, 35, 31, 24, 19, (15)                                                         | 1 2803159225                  |
| 1189h | 34; | 2, 17, 1, 1, 4, (34)                                                                    | 63 43775                      |
|       |     | 30, 1, 34, 15, (2)                                                                      | 11347                         |
| 1189i | 34; | 3, 13, 33, 4, 9, 1, 1, 1, 2, 1, (33)                                                    | 3 90951 47851564921960        |
|       |     | 31, 14, 7, 41, 26, 37, 19, 49, (2)                                                      | —                             |
| 1189j | 34; | 2, 32, 17, 3, 20, 2, 1, 2, 16, 1, 6, 1, 2, (1, 1)                                       | 24 58219 58945                |
|       |     | 33, 5, 3, 23, 36, 25, 4, 57, 9, 45, 20, (33, 33)                                        | 847 64031 17418               |

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines | $y / x$                |
|-------|-----|--------------------------|------------------------|
| 1190  | 34; | (4)                      | 2                      |
|       |     | (34)                     | 69                     |
| 1191  | 34; | 1, 34, (3)               | 3175                   |
| 1192  | —   |                          | 97167                  |
| 1193* | 34; | 1, (22)                  | 33 54723487 3792842959 |

|       | No.                                                                                                                                                                | Continued-fraction lines | $y / x$                             |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------|
| 1194  | —                                                                                                                                                                  |                          | —                                   |
| 1195  | 34; 35, 2, 1, 3, 2, 1, 1, 1, 1, 7, (17)<br>24, 39, 17, 24, 33, 31, 28, 39, 9, (4)                                                                                  |                          | 8549 5380952<br>92268               |
| 1196a | 34; 36, 33, 1, 1, 1, 1, 3, 2, 8, (5, 5)<br>1, —                                                                                                                    |                          | 6264<br>2 16474 678674590           |
| 1196b | 34; 37, 32, 1, 1, 4, 7, 16, 29, 8, (13, 13)<br>9, 1, 1, 1, 6, 3, 1, (10)                                                                                           |                          | 1345 1660293529<br>46500 415        |
| 1196c | 34; 19, 5, 7, 42, 25, 41, 10, 17, 49, (6)                                                                                                                          |                          | 12<br>25                            |
| 1196d | 34; 38, 31, 2, 7, 2, 1, 1, 10, 1, (12)                                                                                                                             |                          | —                                   |
| 1196e | 34; 39, 29, 3, 9, 26, 29, 39, 6, 59, (5)<br>30, (2, (23))                                                                                                          |                          | —<br>—                              |
| 1197  | 40, 28, 27, 16, 1, (8)                                                                                                                                             |                          | 8784589290                          |
| 1198  | —                                                                                                                                                                  |                          | 5116549690                          |
| 1199  | 34; 1, 4, 59, (7)                                                                                                                                                  |                          | 1 21525 49298 36463 39              |
| 1200  | —                                                                                                                                                                  |                          | 42 06256 96350 23592                |
| 1200b | 41, 1, 27, 1, 2, 1, 3, 3, (34)<br>1, 34, 21, 38, 23, 18, 29, 6, 7, 11, 14, 3, 58, 9, 40, 19, 42, 17,<br>21, (2)                                                    |                          | 4917473 75599<br>77820              |
| 1200c | 34; 42, 26, 25, 31, 2, 9, 1, 1, 13, (3)                                                                                                                            |                          | 1351                                |
| 1200d | 43, 35, 25, 7, 34, 35, 5, (22)                                                                                                                                     |                          | —                                   |
| 1200e | 34; 44, 24, (16)                                                                                                                                                   |                          | —                                   |
| 1200f | 34; 1, 2, 23, 19, 9, 4, 4, 2, 1, 1, 1, 7, 13, 1, 2, 1, 2, 1, 1, 4,<br>22, 1, 7, 1,<br>7, 16, 15, 24, 35, 27, 40, 9, 5, 48, 19, 40, 21, 32, 35, 10, 3, 59,<br>8, 45 |                          | —<br>—                              |
| 1201* | 34; 45, 2, 43, (2, (52, 252))                                                                                                                                      |                          | 71 85416 85609 44230                |
| 1202  | —                                                                                                                                                                  |                          | 2490 13832 32746 50571 7474388352   |
| 1203  | 34; 1, 23, (34)                                                                                                                                                    |                          | 10817                               |
| 1204  | 1, 2)                                                                                                                                                              |                          | 312                                 |
| 1205  | —                                                                                                                                                                  |                          | —                                   |
| 1206  | 22, (2) 1, (33)                                                                                                                                                    |                          | —                                   |
| 1207a | 46, 1, 11, —                                                                                                                                                       |                          | 4 3142258212                        |
| 1207b | 34; 1, 2, 2, 57, (2)                                                                                                                                               |                          | 16 95565 54116                      |
| 1207c | 34; 47, 21, 20, 7, 2, 2, 3, 4, 22, 1, (8)<br>29, 9, 27, 25, 19, 16, 3, 60, (7)                                                                                     |                          | 588 33925 37695<br>2 06668<br>74089 |
| 1207d | 48, 20, 16, 1, (12)                                                                                                                                                |                          | 1 454922052<br>33387                |
| 1207e | 34; 1, 24, (63, 14) (5)                                                                                                                                            |                          | 11                                  |
| 1207f | 49, 19, 2, 3, 1, (33)<br>1, 25, (2)                                                                                                                                |                          | —<br>—                              |
| 1207g | 34; 50, 18, 38, 1, 9, 59, (2)                                                                                                                                      |                          | 17611 5293902486                    |
| 1208  | 51, 17, 47, 1, 2, (8)                                                                                                                                              |                          | 42                                  |
| 1209  | 23, 44, 23, (8)                                                                                                                                                    |                          | 4706165263                          |
| 1209b | 34; 52, 16, 1, 3, 2, 1, 1, (22)<br>53, 3, 9, 40, 17, 25, 29, 40, (3)<br>7, —                                                                                       |                          | 29335<br>570<br>—                   |

TABLE 1001 TO 1500—CONTINUED.

|      | No.                                                                   | Continued-fraction lines | $y / x$                  |
|------|-----------------------------------------------------------------------|--------------------------|--------------------------|
| 1210 | 34; 1, 3, 1, 1, 1, 7, 1, 1, (6)<br>54, 15, 39, 26, 41, 9, 6, 31, (10) |                          | 430 99524<br>14992 19281 |
| 1211 | 34; 1, (3)<br>55, (14)                                                |                          | 5<br>174<br>3 65980      |

|       | No. | Continued-fraction lines                                                            | $y \ / \ x$                    |
|-------|-----|-------------------------------------------------------------------------------------|--------------------------------|
|       |     |                                                                                     | 127 41151                      |
| 1212  | 34; | 1, 4, 2, 1, 2, 2, 1, (16)                                                           | 20 27117                       |
|       |     | 56, 13, 24, 37, 23, 21, 47, (4)                                                     | 706 00734                      |
| 1213* | 34; | 57, 12, 49, 13, 33, 36, 7, (27, 27)                                                 | 7321059 0947790823 6704190745  |
|       |     |                                                                                     | 7                              |
| 1214  | 34; | 1, 5, 2, 1, 6, 3, 1, 1, 13, 2, 1, 2, 2, (34)                                        | 244                            |
| 1215  |     | 58, 11, 23, 43, 10, 19, 31, 38, 5, 25, 37, 22, 29, (2)                              | 1916 03685                     |
| 1216  | —   |                                                                                     | 66814 48801                    |
| 1216b | 34; | 1, (5)                                                                              | 7 91969                        |
|       |     | 59, (10)                                                                            | 276 28235469                   |
| 1216c | 34; | 1, 6, 1, 3, 4, 2, 1, 1, 4, (17)                                                     | 10                             |
|       |     | 60, 9, 48, 17, 15, 25, 31, 36, 15, (4)                                              | —                              |
| 1217* | 34; | 1, 7, 1, 2, 1, 3, 1, 1, 1, (1, 1)                                                   | —                              |
| 1218  |     | 61, 8, 47, 19, 43, 16, 37, 29, 32, (31, 31)                                         | —                              |
| 1218b | 34; | 1, (8)                                                                              | —                              |
|       |     | 62, (7)                                                                             | —                              |
| 1219  |     | 1, 10, 1, 1, 1, 7, 9, 1, 5, 2, 4, 5, 6, 1, 3, (1)                                   | 26 76791 47002 595513423815694 |
| 1220  | 34; | 63, 6, 43, 25, 42, 9, 7, 54, 11, 29, 15, 13, 10, 49, 15, (46)                       | 81111 24548                    |
| 1221  | 34; | 64, (5)                                                                             | —                              |
|       |     | 1, 16, (2)                                                                          | 21368152                       |
|       |     | 65, 4, (33)                                                                         | 22 12495 24122                 |
|       |     |                                                                                     | 773 42454 09307                |
| 1222  | 34; | 1, 22, 3, 7, 2, 3, 1, 1, 1, 4, 2, 1, (4)                                            | 122345                         |
| 1223  |     | 66, 3, 22, 9, 29, 17, 38, 27, 39, 14, 23, 42, (13)                                  | 35                             |
|       |     |                                                                                     | 1                              |
| 1223b | 34; | 1, (33)                                                                             | 35                             |
|       |     | 67, (4)                                                                             | 1                              |
| 1224  | 34; | (1) (68)                                                                            | 122365                         |
|       |     |                                                                                     | 44730 59716 99506              |
|       |     |                                                                                     | 15 67486 75428 71047           |
| 1226* | 35; | (7, (01))                                                                           | 21409103                       |
|       |     |                                                                                     | 14                             |
| 1227  | 35; | (35)                                                                                | —                              |
| 1228  | 35; | 23, 2, 1, 7, 8, 1, 1, 1, 2, 2, 1, 1, 5, 3, 1, (16)                                  | 491                            |
|       |     | 3, 24, 43, 9, 8, 41, 27, 36, 23, 24, 31, 37, 12, 17, 51, (4)                        | 584 79034 52350 15227 51177    |
|       |     |                                                                                     | 20517 72574 24010 98134 87200  |
| 1229* | 35; | 17, (1, 1)                                                                          | —                              |
| 1230  |     | 4, (35, 35)                                                                         | —                              |
| 1230b | 35; | (14) (5)                                                                            | —                              |
| 1231  | 35; | 11, 1, 2, 7, 2, 4, 1, 13, 4, 1, 1, 1, 1, 6, 2, 2, 4, 3, 1, 2, 23, (35)              | —                              |
|       |     | 6, 45, 23, 9, 30, 13, 54, 5, 25, 37, 30, 29, 39, 10, 27, 26, 15, 18, 39, 25, 3, (2) | —                              |

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines                                  | $y \ / \ x$            |
|-------|-----|-----------------------------------------------------------|------------------------|
| 1232  | 35; | (1, (07))                                                 | 210 76139 064889938541 |
| 1233  |     | 8, 49, 17, 16, 9, 37, 32, 19, 11, 7, 56, (9)              | 7400 75542 46657       |
| 1234  | —   |                                                           | 1669 10230 73856       |
| 1235  | 35; | 7, 1, 3, 1, 4, 4, 2, 9, 1, 1, 2, 3, 1, 1, (34)            | 58632 78690 67265      |
|       |     | 9, 50, 15, 47, 14, 15, 30, 7, 39, 30, 25, 18, 31, 39, (2) | —                      |
| 1235b | 35; | (7) (10)                                                  | 7                      |
|       |     |                                                           | 246                    |
|       |     |                                                           | 18261 82722            |
|       |     |                                                           | 6 42027 25495          |

|       | No.                                                                                                 | Continued-fraction lines                                | $y \ / \ x$ |
|-------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------|
| 1236  | 35; 6, 2, 1, 1, 1, 4, 1, 3, 1, 1, 2, 1, (22)<br>11, 25, 35, 28, 39, 13, 47, 16, 35, 33, 20, 49, (3) | 30 80599 95913 30265<br>1083 47814 40276 61982          |             |
| 1237* | 35; 5, 1, 5, 1, 1, 3, 1, 1, 2, 23, 17, 1, 1, 5, 2, 1, (7, 7)                                        | 25650                                                   |             |
| 1238  | 12, 51, 11, 36, 33, 17, 36, 31, 27, 3, 4, 37, 33, 12, 23, 44, (9, 9)                                | 9 02501                                                 |             |
| 1239  | 35; 5, 2, 1, 1, (34)<br>13, 26, 29, 41, (2)                                                         | 5<br>176                                                |             |
| 1239b | 35; (5) (14)                                                                                        | 24102<br>8 48719                                        |             |
| 1240  | 35; 4, 1, 2, 7, (2)<br>15, 41, 24, 9, (31)                                                          | 9 65005<br>339 95032                                    |             |
| 1241* | 35; 4, 2, 1, 1, 3, 8, (1, 1)                                                                        | 11780                                                   |             |
| 1242  | 16, 25, 32, 35, 19, 8, (35, 35)                                                                     | 4 15151                                                 |             |
| 1242b | 35; 4, 7, 1, 1, (2)<br>17, 9, 38, 31, (23)                                                          | 4875<br>1 71874                                         |             |
| 1243  | 35; 3, 1, 9, (3)<br>18, 49, 7, (22)                                                                 | 1616 45760 60360<br>57013 08077 08799                   |             |
| 1244  | 35; 19, 40, 23, 20, 11, 8, 43, 25, 44, 5, 31, 13, 55, (4)                                           | 2 13528<br>75 34241                                     |             |
| 1245  | 35; 3, 1, 1, 17, (14)                                                                               | 1257 54312 87288                                        |             |
| 1246  | 20, 31, 39, 4, (5)                                                                                  | 44389 66931 48735                                       |             |
| 1246b | 35; 21, 22, 45, 9, 10, 57, 5, 18, 47, 11, 42, 25, 45, (2)                                           | 26313<br>9 29118589                                     |             |
| 1247  | 35; 3, 5, 9, (1)                                                                                    | 5617                                                    |             |
| 1248  | 22, 13, 7, (58)                                                                                     | 2 61326 40028 30963 92593<br>92 35587 03432 90296 88240 |             |
| 1248b | 35; 23, (4)                                                                                         | 2 757880071<br>21 92943                                 |             |
| 1249* | 35; 2, 1, 13, 2, 7, 2, 1, 2, 3, 1, 3, 1, 1, 1, 4, 1, 3, 1, 8, 23, 2, (4, 4)                         | 775 63250                                               |             |
| 1250* | 24, 45, 5, 32, 9, 25, 37, 24, 17, 45, 16, 39, 27, 40, 13, 48, 15, 57, 8, 3, 31, (15, 15)            | 90969 00792 82080                                       |             |
| 1251  | —                                                                                                   | 32 18812 08291 34849                                    |             |
| 1251b | 35; 2, 1, 4, 2, 1, (1, 1)<br>25, 41, 14, 25, 34, (31, 31)                                           | —<br>—                                                  |             |
| 1251c | 35; 2, 1, 2, 2, 2, 5, (35)<br>26, 37, 23, 25, 27, 13, (2)                                           | —<br>—                                                  |             |
| 1252  | 27, 33, 32, 29, 39, 12, 19, 33, 36, 13, 52, 9, 3, 32, 11, 57, (4)                                   | —                                                       |             |

TABLE 1001 TO 1500—CONTINUED.

|       | No.                                                                                         | Continued-fraction lines | $y \ / \ x$       |
|-------|---------------------------------------------------------------------------------------------|--------------------------|-------------------|
| 1253  | 35; 1, 1, 1, 17, (10)<br>28, 29, 41, 4, (7)                                                 |                          | 78320             |
|       |                                                                                             |                          | 27 72351          |
|       |                                                                                             |                          | 10234             |
|       |                                                                                             |                          | 3 62405           |
| 1254  | 35; 1, 2, 2, 1, (34)<br>29, 25, 11, 49, (2)                                                 |                          | 87 88466 34708    |
|       |                                                                                             |                          | 3113 40025 72889  |
| 1255  | 35; 2, 2, 1, 7, 6, 3, 4, 1, 2, 1, 11, (14)<br>30, 21, 46, 9, 11, 21, 14, 45, 19, 49, 6, (5) |                          | 11075             |
|       |                                                                                             |                          | 3 92499           |
| 1256  | 35; 2, 3, 1, 2, (17)<br>31, 17, 40, 25, (4)                                                 |                          | 2860              |
|       |                                                                                             |                          | 1 01399           |
| 1257  | 35; 2, 4, 1, (22)<br>32, 13, 56, (3)                                                        |                          | 3233              |
|       |                                                                                             |                          | 1 14669           |
|       |                                                                                             |                          | 313 98432 81069   |
|       |                                                                                             |                          | 11140 91438 32810 |
| 1258* | 35; 2, 7, 2, (1, 1)                                                                         |                          | 2                 |

|       | No. | Continued-fraction lines                                                                                                                | $y / x$                              |
|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
|       |     | 33, 9, 26, (33, 33)                                                                                                                     | 694 50461 60725<br>24662 24909 57682 |
| 1259  |     | 35; 1, 13, 1, 2, 3, 2, 1, 1, 5, 1, 6, 4, (35)<br>34, 5, 47, 22, 19, 25, 31, 38, 11, 53, 10, 17, (2)                                     | 17412 0470781<br>6 18556 69263       |
| 1260  |     | 35; (2) (35)                                                                                                                            | 4004<br>1 42297                      |
| 1261* |     | 35; 1, 1, 23, 5, 1, 7, 17, 1, 1, 1, 2, 5, (1, 1)<br>36, 35, 3, 14, 53, 9, 4, 43, 27, 36, 25, 12, (35, 35)                               | 360<br>12799                         |
| 1262  |     | 37, 34, 7, 43, 26, 41, 13, 49, 14, 17, 53, (2)                                                                                          | 2 065989290                          |
| 1263  |     | 35; 1, 1, 5, 1, (22)<br>38, 33, 1, 58, (3)                                                                                              | 149 86708<br>5332 40465              |
| 1264  |     | 35; 1, 1, 4, (4)                                                                                                                        | 175 73127                            |
| 1265  |     | 39, 32, 15, (16)                                                                                                                        | 6255 14462                           |
| 1265b |     | 35; 1, 1, 3, 4, (6)<br>40, 31, 19, 16, (11)                                                                                             | 69835 99490<br>24 86789 07849        |
| 1266  |     | 35; 1, 1, 2, 1, 1, 2, 3, 1, 3, 1, (34)                                                                                                  | 34 299621654                         |
| 1267  |     | 41, 30, 23, 34, 33, 25, 17, 46, 15, 55, (2)                                                                                             | 1 63267 43118<br>58 18371 13691      |
| 1267b |     | 35; 1, 1, 2, 7, 1, 1, 23, (5)<br>42, 29, 27, 9, 34, 37, 3, (11)                                                                         | 32799<br>912007                      |
| 1268  |     | 35; 1, 1, 1, 1, 3, 1, 5, 1, 2, 4, 9, 1, (16)<br>43, 28, 31, 37, 16, 49, 11, 44, 23, 16, 7, 61, (4)                                      | 3<br>—                               |
| 1269  |     | 35; 1, 1, 1, 1, 1, 7, 3, (2)                                                                                                            | 5 14164 62158 51476                  |
| 1270  |     | 44, 27, 35, 28, 41, 9, 20, (27)                                                                                                         | 183 44944 08681 34807                |
| 1270b |     | 35; 1, 1, 1, 3, 1, 1, 1, 1, 5, 1, 7, (14)<br>45, 26, 39, 69, 6, 41, 29, 30, 39, 11, 54, 9, (5)                                          | —<br>—                               |
| 1271  |     | 35; 1, 1, 1, 6, (2)<br>46, 25, 43, 10, (31)                                                                                             | —<br>—                               |
| 1272  |     | 35; 1, (1)                                                                                                                              | —                                    |
| 1273  |     | 47, (24)                                                                                                                                | —                                    |
| 1273b |     | 35; 1, 2, 8, 1, 1, 2, 2, 4, 23, 1, 1, 3, 1, 2, 5, 7, 1, (2)<br>48, 23, 8, 39, 31, 24, 27, 16, 3, 39, 32, 17, 41, 24, 13, 9, 48,<br>(19) | —<br>—                               |

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines                                                   | $y / x$                         |
|-------|-----|----------------------------------------------------------------------------|---------------------------------|
| 1274  |     | 35; 1, 1, 1, 6, 2, 1, 2, (2)                                               | 1571240                         |
| 1275  |     | 49, 21, 49, 10, 25, 38, 23, (26)                                           | [59441 8 004890919              |
| 1276  |     | —                                                                          | 02580                           |
| 1277* |     | 35; 1, 3, (2, (52))                                                        | 3329114 9178452473 36799        |
| 1278  |     | 50, 20, —                                                                  | 931 440647297 42285<br>48482    |
| 1279  |     | —                                                                          | —                               |
| 1280  |     | 35; 1, 1, 2, 2, 2, 7, 1, 1, 9, 1, 2, 13, 1, (16)                           | —                               |
| 1281  |     | 51, —                                                                      | —                               |
| 1282  |     | 33, 24, 25, 28, 9, 35, 36, 7, 45, 24, 5, 63, (4)                           | —                               |
| 1283  |     | —                                                                          | —                               |
| 1284  |     | 35; 3, 2, 5, 17, 1, 2, (6, 6)<br>17, 29, 13, 4, 47, 23, (11, 11)           | —<br>—                          |
| 1285* |     | —                                                                          | 4                               |
| 1286  |     | —                                                                          | 143                             |
| 1287  |     | 52, 44, (2)                                                                | —                               |
| 1288  |     | —                                                                          | —                               |
| 1289* |     | 53, (18)                                                                   | —                               |
| 1290  |     | 35; 17, 15, 1, 1, 13, 1, 3, 23, 1, 1, 2, 2, 1, 5, 1, 3, 1, 11, 7, 3,<br>44 | 7 89404 78828 27783 39935 75041 |

|       | No.                                                                                                                                                                                                              | Continued-fraction lines                                                               | $y \ / \ x$ |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------|
| 1291  | 1, 33, 38, 19, 15, 51, 18, 3, 41, 30, 25, 22, 45, 11, 50, 15, 53, 6, 9, 55, 16, 37, 1,                                                                                                                           | 282 31569 87094 18598 91612 561040409                                                  |             |
| 1292  | 54, 1, 1, 1, 2, (35)<br>1, 1, 30, 33, 31, 34, 27, (2)                                                                                                                                                            | —<br>—                                                                                 |             |
| 1292b | 35; (1, (74))<br>55, —                                                                                                                                                                                           | 220 51841<br>81748                                                                     |             |
| 1292c | 35; 3, 1, 2, 13, 1, (22)<br>1, 3, 1, 16, 41, 25, 5, 64, (3)                                                                                                                                                      | 29175<br>—                                                                             |             |
| 1292d | 56, 47, 1, 3, 9, 1, (34)<br>3, 9, 49, 18, 7, 63, (2)                                                                                                                                                             | 7903 2467410338<br>34103                                                               |             |
| 1292e | 35; 57, 13, 1, 4, 1, 9, 1, 2, 4, 1, 3, 2, 2, (35)<br>1, 37, 7, 46, 23, 14, 47, 17, 26, 29, (2)                                                                                                                   | 5 15591522 61505<br>65140 88226                                                        |             |
| 1292f | 35; 58, 4, —<br>65, —                                                                                                                                                                                            | 184<br>24                                                                              |             |
| 1292g | 19, 1, 1, 6, 24, 34, —<br>11, —                                                                                                                                                                                  | 2156<br>877                                                                            |             |
| 1292h | 35; 1, 7, 2, 2, 1, 17, 4, 1, 2, 1, (1, 1)<br>10, 3, 53, 36, 9, 29, 21, 49, 4, 15, 44, 21, 36, (31, 31)                                                                                                           | 48426<br>68507 94225<br>5674372178887                                                  |             |
| 1292i | 35; 5, 61, 2, 1, 2, 13, 1, (34)<br>62, 25, 37, 26, 5, 65, (2)                                                                                                                                                    | 2 53631 10565<br>7072<br>8                                                             |             |
| 1292j | 35; (9) 5,<br>63, ((76))                                                                                                                                                                                         | 9<br>323                                                                               |             |
| 1292k | 35; (8) 40, 19, 2, 8, 1, 1, 1, 1, 2, 1, 1, 13, 1, 3, (1, 1)<br>64, 25, 8, 41, 29, 32, 35, 23, 31, 40, 5, 53, 16, (35, 35)                                                                                        | 835 94139 2313980530<br>30012 50804                                                    |             |
| 1292l | 35; 65, 7, 1 (6)                                                                                                                                                                                                 | 12<br>431                                                                              |             |
| 1292m | 35; 66, (10) 1, 1, 2, 3, 1, 1, 1, 1, 4, 1, 11, 6, 2, 4, 3, 23, 1, 1, 15, 33, 34, 25, 18, 37, 31, 30, 39, 13, 55, 6, 11, 30, 15, 22, 3, 45, 26,<br>1, 7, 1, 3, 2, 1, 6, 2, (35)<br>9, 50, 17, 23, 45, 10, 33, (2) | 94899 51267<br>74456<br><br>1974 27745 80291 79860<br>70936 75339 57910 18687<br>08410 |             |
| 1292n | 35; (16) 4, (4)                                                                                                                                                                                                  | 64178<br>41                                                                            |             |

TABLE 1001 TO 1500—CONTINUED.

|       | No.                                                                         | Continued-fraction lines      | $y \ / \ x$ |
|-------|-----------------------------------------------------------------------------|-------------------------------|-------------|
| 1293  | 35; 1, (22)<br>68, (3)                                                      | 24<br>863                     |             |
| 1294  | 35; 1, (34)<br>69, (2)                                                      | 1295<br>36                    |             |
| 1295  | 35; (1) (70)                                                                | 1<br>36                       |             |
| 1297* | 36; (7, (20))                                                               | 129337661<br>24               |             |
| 1298  | 36; (36) (24)                                                               | —                             |             |
| 1299  | 36; (3)                                                                     | 86158<br>649                  |             |
| 1300  | 36; (18) (4)                                                                | —                             |             |
| 1301* | 36; 14, 2, (2, 2)<br>5, 29, (25, 25)                                        | 6025<br>2 17318               |             |
| 1302  | 36; (12) (6)                                                                | 43132                         |             |
| 1303  | 36; 10, 3, 2, 1, 23, 2, 1, 2, 1, 3, 3, 1, 1, 7, 2, 5, 11, 1, 5, 1, 1, 1, 4, | 10127 26314 51893 46887 71461 |             |

|       | No. | Continued-fraction lines                                                                                                | $y / x$                                   |
|-------|-----|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
|       |     | 7, 21, 22, 49, 3, 26, 39, 21, 43, 18, 19, 33, 38, 9, 31, 13, 6, 57,<br>11, 42, 27, 41, 14,<br>1, 1, (38)<br>33, 39, (2) | 3 65564 74232 60608 71132 28808<br>—<br>— |
| 1304  |     | 36; (9) (8)                                                                                                             | 2899<br>325                               |
| 1305  |     | 36; (8) (9)                                                                                                             | 8                                         |
| 1306* |     | 36; 7, 4, 1, 2, 11, 1, 2, 4, 2, 9, 1, (7, 7)                                                                            | 862 68151 05389                           |
| 1307  |     | 10, 15, 42, 25, 6, 47, 23, 15, 31, 7, 58, (9, 9)                                                                        | 31176 12101 50515                         |
| 1308  |     | 36; 11, 37, 34, 17, 43, 22, 31, 41, (2)                                                                                 | 1957 48028127<br>6                        |
| 1308b |     | 36; (6) (12)                                                                                                            | —                                         |
| 1309  |     | 13, 36, 35, 15, 19, 36, 33, 25, 21, 4, 55, 15, 52, 9, (17)                                                              | 991 09921 7 99544<br>80317 01740 72265    |
| 1310  |     | 36; 5, 6, 2, 1, 1, 1, 1, 1, (6)<br>14, 11, 26, 35, 31, 34, 29, 41, (10)                                                 | 34 56382<br>1251 00021                    |
| 1311  |     | 36; 4, 1, 4, 2, (1)                                                                                                     | 17                                        |
| 1312  |     | 15, 49, 14, 25, (38)                                                                                                    | 5353<br>1 931842607                       |
| 1312b |     | 36; 4, 1, 1, (17)<br>16, 33, 39, (4)                                                                                    | 53137<br>616                              |
| 1313* |     | 36; (4, 4) (17, 17)                                                                                                     | 58                                        |

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines                                                                                                                                                               | $y / x$                                                                                                                                         |
|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1314  |     | (4) (18)                                                                                                                                                                               | 1454<br>924448                                                                                                                                  |
| 1315  |     | 36; 20, 46, 15, 6, 59, 9, (10) (2)                                                                                                                                                     | 8875 7141365416                                                                                                                                 |
| 1316  |     | 1, 1, 1, 1, 2, 3,                                                                                                                                                                      | —                                                                                                                                               |
| 1317  |     | 21, 37, 31, 52, 35, 25, 19, (28)                                                                                                                                                       | 65508                                                                                                                                           |
| 1318  |     | 2, 3, 1, 5, 3, 1, 1, 1, 4, 1, 17, 3, 10, (24)                                                                                                                                          | 23                                                                                                                                              |
| 1319  |     | 3, 28, 17, 49, 12, 19, 39, 28, 41, 13, 57, 4, 23, 7, (3)                                                                                                                               | 8716540105                                                                                                                                      |
| 1319b |     | 22, 19, 31, 36, (5, 43) 17, 21, 14, 53, 9, 18, 51, 7, 42, 29, 33,<br>34, 27, 6, 49, 21, (2)<br><br>38, 11, (2)<br><br>23, 1, 6, 1, 13, 1, 1, 1, (35)<br><br>10, 59, 5, 46, 25, 47, (2) | 1 03450 99306<br><br>37 54287 70770<br>93751<br>3138 36679 69954 08957<br>1 13936 11468 26276 81762 974655147<br>334<br>12130 01011<br>59214009 |
| 1320  |     | 36; (23, (43))                                                                                                                                                                         | 3                                                                                                                                               |
| 1321* |     | 25, 1, 1, 2, 1, 8, 2, 2, 1, 1, 3, 1, 23, 2, 4, 2, 1, 4, 6, 2, 1, 1, 7,<br>2, (14, 14)                                                                                                  | 14 70631 38627 6169162171 69925                                                                                                                 |
| 1322* |     | 26, 45, 8, 29, 24, 33, 37, 16, 55, 3, 32, 15, 24, 43, 15, 11, 27,<br>31, 40, 9, 33, (5, 5)                                                                                             | 534 50926 71539 96432                                                                                                                           |
| 1323  |     | —                                                                                                                                                                                      | 29885211                                                                                                                                        |
| 1324  |     | 36; 1, 3, 1, (1, 1)<br>41, 17, 38, (31, 31)                                                                                                                                            | —<br>—                                                                                                                                          |
| 1325* |     | 27, 1, 2, 7, 1, 2, (2)                                                                                                                                                                 | 1 00360                                                                                                                                         |
| 1326  |     | 37, 26, 9, 47, 22, (27)                                                                                                                                                                | 50101                                                                                                                                           |
| 1327  |     | —                                                                                                                                                                                      | —                                                                                                                                               |
| 1328  |     | 28, —                                                                                                                                                                                  | —                                                                                                                                               |
| 1329  |     | —                                                                                                                                                                                      | —                                                                                                                                               |
| 1330  |     | 1, 1, 2, 2, 3, 4, 1, 1, 3, 1, 2, 1, 2, 5, 1, 2, 3, 8, 1, 3,                                                                                                                            | 36 21799                                                                                                                                        |
| 1331  |     | 33, 35, 24, 27, 20, 15, 36, 35, 17, 44, 21, 40, 25, 12, 45, 23,<br>21, 8, 53, 15, 56, 15, 51902,                                                                                       | 12718 55353 72450 38443 74810                                                                                                                   |

|       | No.                                                                                                                                       | Continued-fraction lines                     | $y \ / \ x$ |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------|
| 1332  | 2, 23, 2, 6, 7, 1, 13, 1, 2, 10, (18)<br>29, 3, 33, 11, 9, 60, 5, 47, 24, 7, (4)                                                          | 42650 10838 17312 41819 40987<br>—           |             |
| 1332b | 36; (29) (2)<br><br>2, 25, (26)                                                                                                           | —<br>5<br>182<br>2549                        |             |
| 1332c | 36; 30, 2, 1, (2, (35))<br>21, 51, (2)                                                                                                    | 178750<br>65032649<br>13447                  |             |
| 1332d | 3, 1, (3)<br>17, 47, (16)                                                                                                                 | —<br>—                                       |             |
| 1332e | 36; 32, 5, 8, 1, 13, 1, 2, 4, 4, 1, 1, 1, 2, 2, 1, 1, 6, (24)<br>13, 8, 61, 5, 34, 18, 23, 16, 15, 40, 29, 37, 24, 25, 32, 39, 11,<br>(3) | 19 32883 59509 0314419156<br>704 64145 63882 |             |
| 1332f | 36; 33, 7, 1, 1, 1, 1, (4)<br>9, 41, 30, 31, 39, (14)                                                                                     | 344615829<br>12<br>10727                     |             |
| 1332g | 34, 14, 10, 2, 1, 4, 1, 1, 6, 1, 2, 1, (35)<br>5, 7, 25, 43, 14, 35, 37, 10, 49, 19, 53, (2)<br>35, —<br>(36)                             | 273 0230738710 19890<br>—<br>9961<br>73      |             |

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines                                         | $y \ / \ x$                |
|-------|-----|------------------------------------------------------------------|----------------------------|
| 1333  | 36; | 1, 1, 1, 23, 1, 1, 5, 7, 1, 17, 2, 1, 1, 1, 5, (2)               | 904 20166 84140            |
| 1334  | 37, | 36, 1, 59, 12, 9, 61, 4, 27, 36, 29, 41, 12, (31)                | 33012 64933 32151          |
| 1335  | 36; | 13, 1, 2, 2, 1, 1, (2)                                           | 31585694 08256             |
| 1336  | 5,  | 49, 22, 25, 34, 35, (23)                                         | 721029355                  |
| 1337  | 38, | 35, 1, 79, (7)                                                   | —                          |
| 1338  | —   |                                                                  | —                          |
| 1339  | —   |                                                                  | —                          |
| 1340  | —   |                                                                  | —                          |
| 1341  | 39, | 34, 15, (10)                                                     | 45124                      |
| 1342  | 19, | 25, 39, 24, 23, 44, 15, 12, 55, 9, 20, 33, 39, (8)               | 174 56099                  |
| 1343  | —   |                                                                  | 6380 43737 9199662925      |
| 1344  | 40, | 33, 2, 1, 8, 2, 4, (10)                                          | 373 31647                  |
| 1345  | 11, | —                                                                | 13674 97272                |
| 1346  | 41, | 32, 23, 31, 23, 2, 47, 8, 31, 16, (7)                            | 4                          |
| 1347  | 1,  | 2, 10, (12)                                                      | 173                        |
| 1348  | —   |                                                                  | 73194                      |
| 1349  | 42, | 31, 27, 39, 26, 7, (6)                                           | 08813                      |
| 1350  | 4,  | 1, (4)                                                           | 4104                       |
|       | 14, | 51, (13)                                                         | 1 50175                    |
| 1351  | 36; | 43, 23, 19, 1, 11, 61, (18)                                      | 7 19932                    |
|       | 40, | (4)                                                              | 4259362310                 |
| 1352  | 44, | 28, 11, 31, 1, 1, 2, 3, 3, 1, 1, 3, 1, 2, 1, 7, 2, 2, 17, 1, 10, | 36359 14354 86763          |
|       | 1,  | 1, 14, (8)                                                       |                            |
| 1353  | 35, | 31, 36, 25, 20, 19, 35, 36, 17, 45, 20, 49, 9, 28, 29, 4, 63,    | 13 31459 72360 20892 39201 |
|       | 7,  | 36, 37, 5, (9)                                                   |                            |
|       | 45, | 27, —                                                            | 109390                     |
| 1353b | 36; | 46, 1, 26, 39, (22)                                              | 8180152981                 |
|       | (2) |                                                                  | 165                        |
| 1353c | 4,  | 1, (35)                                                          | 3                          |
| 1353d | 47, | 25, 43, 2, 13, 59, (2)                                           | —                          |
| 1353e | 36; | (17)                                                             | —                          |
|       | 48, | 24, 47, (4)                                                      | 6049                       |

|       | No.  | Continued-fraction lines                                                                                                                             | $y / x$                                                              |
|-------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|       | (14) |                                                                                                                                                      | 132<br>4841                                                          |
| 1353f | 36;  | 49, 23, 1, 24, 2, 1, 9, 1, 2, (36)<br>12, 1, 2, 55, 7, 46, 25, (2)                                                                                   | 3405 82362<br>60 50115                                               |
| 1353g | 36;  | 50, 22, 23, 4, 2, 1, 5, 1, (35)<br>46, 1, 11, 61, (2)                                                                                                | 1 64753<br>92 8426367862<br>56 21145 67828 13793<br>2063 81035 31297 |
| 1353h | 51,  | 21, 2, 20, 1, 23, 1, 9, 1, 1, 7, 1, 1, 1, 2, 1, 5, 2, 1, 1, 5,<br>(18)<br>1, 41, 3, 64, 7, 37, 36, 9, 43, 28, 39, 21, 47, 12, 27, 32, 39, 13,<br>(4) | 5 55200 34840<br>203 91806 65951                                     |
| 1353i | 36;  | 52, 32, —                                                                                                                                            | 12804<br>4 70414698                                                  |
| 1353j | 36;  | 2, 5, 3, 2, 17, 1, 13, 1, (2)<br>53, 19, 41, 25, 13, 20, 31, 4, 65, 5, 52, (19)                                                                      | 6175<br>3107<br>1 14243                                              |
| 1353k | 36;  | 1, 7, 2, (2)<br>54, 18, 50, 9, 2, 29, (25)                                                                                                           | 42191<br>8628453336                                                  |
| 1353l | 36;  | (10)<br>55, 3, 17, 1, 23, (7) 1, (17)                                                                                                                | —<br>—                                                               |
| 1353m | 36;  | 56, 16, 3, 49, (4)                                                                                                                                   | —                                                                    |
| 1353n | 36;  | 1, 1, 1, 2, 1, 8, (2)<br>31, 32, 37, 21, 49, 8, (33)<br>57, 3, 39, —                                                                                 | —<br>—<br>—                                                          |

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines                                                                                                                                            | $y / x$                       |
|-------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 1354* | 36; | 58, 14, 1, 5, 15, 1, 2, 2, 6, 1, 7, 3, 4, 1, 1, (2, 2)                                                                                                              | 131355 7459053930 9948611635  |
| 1355  | 59, | 33, 10, 57, 9, 22, 15, 38, 33, (25, 25)                                                                                                                             | 918484293523829               |
| 1356  | —   |                                                                                                                                                                     | —                             |
| 1357  | 2,  | (14)                                                                                                                                                                | 4918739790                    |
| 1358  | 25, | 26, (5)                                                                                                                                                             | 5212                          |
| 1359  | —   |                                                                                                                                                                     | 1 9196229                     |
| 1360  | —   |                                                                                                                                                                     | —                             |
| 1361* | 36; | 60, 13, 1, 41, 8, 2, 1,                                                                                                                                             | 9 65303 46430 64640           |
| 1362  | 1,  | 11, —                                                                                                                                                               | 355 59347 94143 8721345014    |
| 1363  | 5,  | 1, 4, (18)                                                                                                                                                          | —                             |
| 1364  | —   |                                                                                                                                                                     | —                             |
| 1365  | 61, | 12, 44, 8, 57, 11, 52, 15, (4)<br>1, 6, 1, 7, 1, 1, 1, 5, 2, 17, 1, 23, 1, 1, 1, 1, (2)<br>62, 11, 22, 37, 9, 44, 27, 43, 12, 33, 4, 67, 3, 44, 29, 33, 36,<br>(23) | —<br>—<br>—                   |
| 1365b | 36; | 1, 36, 2, (36)                                                                                                                                                      | —                             |
| 1365c | 36; | 63, 10, 27, 47, 51, 15, 8, 34, 35, 25, 23, 45, 14, 37, 35,<br>(18)                                                                                                  | 468 99823<br>17281 4788023410 |
| 1365d | 36; | 64, 1, 1, 17, (3)                                                                                                                                                   | 85769<br>2 7749               |
| 1365e | 36; | 65, 7, 4, 41, 9, 6, (16)                                                                                                                                            | 58529                         |
| 1365f | 36; | 9, 1, 4, 2, 1, 2, 1, 1, 14, 5, 1, (1, 1)<br>25, 40, 23, 32, 41, 5, 13, 40, (31, 31)                                                                                 | 35 93384<br>1325 66186 45260  |
| 1365g | 36; | 66, 11, 6, 4, 1, 23, 3, 1, 4, 2, (36)                                                                                                                               | 13730711 2847205478           |
| 1365h | 36; | 67, 7, 39, 17, 49, 14, 33, (2)                                                                                                                                      | —                             |
| 1365i | 36; | 1, 9, 1, 1, 3, 34, (1) 1,<br>68, 17, 16, (58) 1, 2, (6)                                                                                                             | 9531479265<br>1062 2867575310 |

|                       | No.                                                                                                                                  | Continued-fraction lines | $y / x$                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------|
|                       | 7, 43, 25, (11)                                                                                                                      |                          | 34                                                                                     |
| 1365j                 | 69, 31, —<br>35, 55, 9, (35) 6, 4                                                                                                    |                          | 684<br>25271                                                                           |
| 1365k<br>1366         | 36; 4, 10, 1, 7, 1, 1, 4, 2, 1, 1, 14, 5, 4, 1, 2, 1, 2, 2,<br>70, 5, 9, 38, 35, 15, 27, 31, 42, 5, 14, 15, 46, 21, 42, 23, 30,<br>— |                          | —<br>61 98787 91120 98468 23128 64853 37<br>2291 03705 28461 89335 14238 95408 6404327 |
| 1367<br>1368          | 3, 17, 1, 3, 2, 1, 1, 1, 3, 3, 1, 4, 1, 11, 2, (36)<br>1, 25, 37, 30, 39, 19, 18, 49, 13, 57, 6, 35, (2)                             |                          | —<br>—                                                                                 |
| 1370*<br>1371<br>1372 | 36; (35)                                                                                                                             |                          | 99525<br>1368<br>37<br>1                                                               |
| 1373*                 | 36; (1) (72)                                                                                                                         |                          | 371                                                                                    |
|                       | 37; (74)                                                                                                                             |                          | 1370                                                                                   |
|                       | 37; (37) 24, 24,                                                                                                                     |                          | —                                                                                      |
|                       | 37; 1, 1, 8, 2, 1, 1, 1, 2, 2, 5, 1, 3, (18)<br>7, 19, 58, 27, 36, 31, 37, 24, 29, 12, 49, 19, (4)<br>49, —                          |                          | 92062 30469 2374156527<br>34 10035 48679<br>685                                        |
|                       | 37; 14, 43, (37, 37) (21)                                                                                                            |                          | 25382                                                                                  |
|                       | 37; 4, 2, 1, 3, 4, 1, 2, (24)<br>57, 9, 42, 19, 15, 43, 26, (3)<br>5, — (1 ...)                                                      |                          | 41422 8295007445<br>15 35443<br>—                                                      |

TABLE 1001 TO 1500—CONTINUED.

|                       | No.                                                                                                                                                   | Continued-fraction lines | $y / x$                                                                          |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------|
| 1375<br>1376          | 37; 12, 2, 1, 7, 1, 1, 3, 2, 1, 2, (6)<br>6, 25, 46, 9, 39, 34, 19, 25, 39, 26, (11)                                                                  |                          | 24090 73932<br>8 93308 52249                                                     |
| 1376b                 | 37; 10, 1, 1, 2, 2, 3, 1, (17)<br>7, 41, 32, 25, 28, 17, 55, (4)                                                                                      |                          | 58 37205<br>2165 28049<br>11121 62480<br>4 12700 701410513                       |
| 1377<br>1378*         | 37; 9, 3, 1, 3, 1, 7, 2, 5, (4)<br>8, 19, 47, 16, 53, 9, 32, 13, (17)                                                                                 |                          | 42801<br>678 06016<br>25179 68895                                                |
| 1378b                 | 37; 8, (4, 4)<br>9, (17, 17)                                                                                                                          |                          | 182<br>6761                                                                      |
| 1379                  | 37; 7, 2, 2, 2, 1, 1, 3, 3, (10)<br>10, 29, 26, 25, 34, 37, 19, 22, (7)                                                                               |                          | —<br>—                                                                           |
| 1380                  | 37; 6, 1, (2)<br>11, 49, (20)                                                                                                                         |                          | —<br>—                                                                           |
| 1380b                 | 37; 6, 5, 1, 1, 4, 2, 2, 3, 3, 4, 14, 1, 1, 1, 2, 1, 1, 2, 1, 24,<br>12, 13, 37, 36, 15, 28, 27, 20, 21, 17, 5, 45, 28, 39, 23, 35, 36,<br>21, 52, 3, |                          | —<br>—                                                                           |
| 1381*                 | 18, 1, 1, 5, 1, 2, 7, 1, 9, 1, 2, 1, (4, 4)<br>4, 39, 35, 12, 45, 25, 9, 60, 7, 51, 20, 47, (15, 15)                                                  |                          | 5 76794 95050 76394 67609 26124 87905<br>214 34743 40923 77992 50381 97708 76282 |
| 1382                  | 37; 5, 1, 2, 2, 1, 1, (36)<br>13, 46, 23, 26, 31, 43, (2)                                                                                             |                          | 1303 4039278327<br>12 9344984426815                                              |
| 1383                  | 37; 5, 3, 2, 1, (11)<br>14, 21, 23, 49, (6)                                                                                                           |                          | 17299<br>2797<br>1 04092                                                         |
| 1384<br>1385*<br>1386 | 37; 4, 1, (17)<br>15, 57, (4)<br>—                                                                                                                    |                          | 21259752<br>40557<br>15 10442                                                    |
| 1386b                 | 37; 4, 1, 1, 1, (3, 3)<br>16, 41, 29, 40, (19, 19)                                                                                                    |                          | 2 20986 97686<br>82 33062 52807                                                  |
| 1386c                 | 37; 4, 2, 1, (2)                                                                                                                                      |                          | 16796                                                                            |

|       | No. | Continued-fraction lines                                           | $y / x$                                            |
|-------|-----|--------------------------------------------------------------------|----------------------------------------------------|
|       |     | 17, 25, 41, (22)                                                   | 6 25975<br>6165 38516 07582<br>2 29862 22854 32981 |
| 1387  |     | 37; 4, 8, (37)<br>18, 9, (2)                                       | 90 81305<br>156472                                 |
| 1388  |     | 37; 3, 1, 9, 1, 8, 2, 2, 5, 3, (18)                                | —                                                  |
| 1389  |     | 19, 52, 7, 61, 8, 29, 28, 13, 23, (4)                              | —                                                  |
| 1389b |     | 37; 3, 1, 2, 2, (24)<br>20, 43, 23, 31, (3)                        | —<br>—                                             |
| 1390  |     | 37; 3, 1, 1, 6, 4, 1, 4, 1, 1, 11, 1, 7, 2, 1, 2, 1, (6)           | —                                                  |
| 1391  |     | 21, 34, 39, 11, 15, 51, 14, 35, 39, 6, 61, 9, 26, 41, 21, 49, (10) | —                                                  |
| 1392  |     | 12, 25, 38, 29, 43, 10, 55, (13)                                   | —                                                  |
| 1392b |     | 37; 3, (4)<br>23, (16)                                             | —<br>—                                             |
| 1393  |     | 37; (34)                                                           | 96                                                 |
| 1394  |     | (1, (07))<br>2, 1, (36)<br>25, 49, (2)                             | 3538336<br>12545<br>—                              |

TABLE 1001 TO 1500—CONTINUED.

|       | No. | Continued-fraction lines                                                                                                                                                                                                                                   | $y / x$                                                                                                      |
|-------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1395  |     | 37; 26, 2, 11, 1, (1, (89))                                                                                                                                                                                                                                | 45 3320                                                                                                      |
| 1396  |     | —                                                                                                                                                                                                                                                          | 1                                                                                                            |
| 1397  |     | 20, 1, —                                                                                                                                                                                                                                                   | 24001                                                                                                        |
| 1398  |     | —                                                                                                                                                                                                                                                          | —                                                                                                            |
| 1399  |     | 27, 45, 3, 5, 48, 25, (1, (84))                                                                                                                                                                                                                            | 169964840530                                                                                                 |
| 1400  |     | —                                                                                                                                                                                                                                                          | —                                                                                                            |
| 1401  |     | 28, 37, 2, 1, 29, 14, 1, —                                                                                                                                                                                                                                 | 4289 48201                                                                                                   |
| 1402* |     | 37; 2, 37, 24, 41, 6, 17, 18, 1, 1, 2, (6)                                                                                                                                                                                                                 | 66440                                                                                                        |
| 1403  |     | 4, 43, 31, 28, (11)                                                                                                                                                                                                                                        | —                                                                                                            |
| 1404  |     | 37; 1, (36)                                                                                                                                                                                                                                                | 1 6033428 48351                                                                                              |
| 1405  |     | —                                                                                                                                                                                                                                                          | —                                                                                                            |
| 1406  |     | 12, 3, 2, 51, (2)                                                                                                                                                                                                                                          | 1 3606145347                                                                                                 |
| 1407  |     | —                                                                                                                                                                                                                                                          | —                                                                                                            |
|       |     | 23, 38, 14, 1, 7, 2, 1, 1, 1, 4, 2, 1, 3, 2, 7, 24,<br>—<br>37; 2, 1, 4, (37)                                                                                                                                                                              | 19 32130 14430 04420 15349<br>449<br>722 67866 47899 40453 09640                                             |
|       |     | 30, 11, 14, 5, 62, 9, 27, 37, 30, 41, 15, 25, 43, 18, 31, 10, 3,<br>58, 15, (2)<br>37; 2, —<br>(25)<br>37; 2, 21, 1, 2, 5, 2, 2, 1, 1, 6, 4, 1, 1, 8, 1, 4, 10, 2, (24)<br>32, 17, 5, 49, 24, 13, 29, 25, 33, 40, 11, 16, 35, 39, 8, 55, 15,<br>7, 35, (3) | —<br>12<br>—<br>44183 34363 33784<br>16 53779 66015 12296 9857091352<br>2 48379 80029 00509<br>93 00157 1995 |
|       |     | 37; 2, 3, 11, 1, 1, 2, 1, 7, 1, (1, 1)<br>79, 63, 1, 6, 41, 33, 22, 49, 9, 42, (31, 31)<br>33, 13, 19, 3, 54, 5, (1)                                                                                                                                       | —<br>—<br>—                                                                                                  |
|       |     | 34, 3, (46)                                                                                                                                                                                                                                                | 74176266<br>62425                                                                                            |
|       |     | 37; 2, 7, 1, (4)<br>35, 9, 56, (13)                                                                                                                                                                                                                        | —<br>2249                                                                                                    |
|       |     | 37; 2, (1, (45)) 38, 36,                                                                                                                                                                                                                                   | 602<br>100<br>01357257915                                                                                    |

|       | No.                                                            | Continued-fraction lines            | $y \ / \ x$           |
|-------|----------------------------------------------------------------|-------------------------------------|-----------------------|
|       | 37;                                                            | (37)                                | 3211                  |
|       |                                                                |                                     | 1 20487 4967231777    |
|       | 37;                                                            | 1, (26, (43)) 37, 1,                | 94474 40193 53380     |
|       |                                                                |                                     | 35 46252 44206        |
|       | 16, 39, 1, 2, 7, 1, (17)                                       |                                     | 20                    |
| 1408  | —                                                              |                                     | 118 21953             |
| 1409* | 11, 33, 28, 9, 63, (4)                                         |                                     | 4440                  |
| 1410  | —                                                              |                                     | 71373501              |
| 1411  | 39, 36, 7, 3, 9,                                               |                                     | 1 012577013 014535932 |
| 1412  | —                                                              |                                     | 124 23214 3310874490  |
| 1413  | 23, 8, —                                                       |                                     | 4669 87287            |
| 1414  | —                                                              |                                     | —                     |
| 1415  | 40, 35, 5, 64, 7, 55, 16, 49, 17, 40, 31, 35, 32, 37, (25, 25) |                                     | 485 26483 4121289190  |
|       | 37;                                                            | 1, (4)                              | 18247 50630           |
|       | (19, 1, 1, 2)                                                  |                                     | 2205 2895060241       |
|       | 41, 34, 33, 23, 30, 15, 38, 35, 21, (2)                        |                                     | —                     |
|       | 37;                                                            | 42, 1, 1, 1, 3, 4, 6, 1, 1, 2, (18) | —                     |
|       | 43, 32, 1, 1, 44, 19, 17, 11, 41, 32, 29, (4)                  |                                     | —                     |
|       | 44, 1, 3, 1, 6, 18, 1, 1, 1, 5, (8)                            |                                     | —                     |
|       | 19, 36, 37, 17, 52, 11, 4, 47, 27, 44, 13, (9)                 |                                     | —                     |
|       | 45, 29, 30, 31, 1, 11, 1, 14, 8, 3, 2, 5, 2, 1, (4)            |                                     | —                     |
|       | 43, 6, 65, 5, 9, 21, 30, 13, 25, 45, (14)                      |                                     | —                     |
|       | 41, 14, 49, 19, (10)                                           |                                     | —                     |
|       | 46, 35, 34, 31, —                                              |                                     | —                     |

TABLE 1001 TO 1500—CONTINUED.

|       | No.                                                                                         | Continued-fraction lines     | $y \ / \ x$                      |
|-------|---------------------------------------------------------------------------------------------|------------------------------|----------------------------------|
| 1416  | 37;                                                                                         | 1, 28, 25, 2, 1, (8)         | 6858                             |
| 1417* | 47, 1, 39, 16, 23, 49, (8)                                                                  |                              | 2 58065                          |
|       | 1, 27, 24, 1, 7, (2, 2)                                                                     |                              | 435                              |
|       | 48, 43, 3, 64, 9, (29, 29)                                                                  |                              | 72829                            |
|       | 26, 47, 2, (61, (07, 7))                                                                    |                              | 16402 14636                      |
| 1418* | —                                                                                           |                              | 381803933939                     |
| 1419  | 49, 25, —                                                                                   |                              | —                                |
| 1420  | (3, (72))                                                                                   |                              | —                                |
| 1421  | —                                                                                           |                              | 12770                            |
| 1422  | —                                                                                           |                              | —                                |
| 1423  | 37;                                                                                         | 50, 24, 12, 22, 36, 39, (14) | 2554383089                       |
| 1424  | —                                                                                           |                              | 9 260                            |
| 1425  | 37;                                                                                         | 51, 23, —                    | —                                |
| 1426  | —                                                                                           |                              | 31                               |
| 1427  | (2, (92))                                                                                   |                              | 1199                             |
| 1428  | —                                                                                           |                              | —                                |
|       | 52, 2, 18, 29, —                                                                            |                              | 9801                             |
| 1429* | —                                                                                           |                              | 08097                            |
| 1430  | 37;                                                                                         | 1, 1, 1, 3, 5, (8)           | 79792                            |
| 1431  | —                                                                                           |                              | 127 24746 01715 56263 0946270589 |
| 1432  | 3, 47, 19, 14, (9)                                                                          |                              | 4800 11423 18614 76037           |
|       | 53, —                                                                                       |                              | 13250                            |
| 1433* | —                                                                                           |                              | 5 00001                          |
| 1434  | 21, —                                                                                       |                              | 1514                             |
|       | 54, 20, 17, 38, 33, 17, 31, 42, 11, 18, 3, 66, 7, 57, 14, 33, 6, 9, 22, 21, 19, 42, 27, (2) |                              | —                                |
|       | 37;                                                                                         | 1, 2, (4)                    | —                                |
|       | 1, 44, 25, (16)                                                                             |                              | —                                |
|       | 55, 47, 3, —                                                                                |                              | —                                |

| No. Continued-fraction lines |                                                    | $y / x$                  |
|------------------------------|----------------------------------------------------|--------------------------|
|                              | 37; (2) (19)                                       | —                        |
|                              | 37; 56, 18, 15, 2, 3, 1, 1, 1, (2)                 | 2 601917352              |
|                              |                                                    | 98 47097                 |
|                              | 37; 57, 17, 41, 13, 17, 41, 50, 39, (23)           | 2481                     |
|                              |                                                    | 33                       |
|                              |                                                    | 93739 18762              |
|                              | 58, 3, 16, 31, 26, 41, 23, 37, 34, 29, (2)         | 340910                   |
|                              | 37; 1, (2)                                         | 89305                    |
|                              | 65, 9, 13, 5, 14, 87, (212)                        | 75918                    |
|                              | 37; (2, (32, 232))                                 | 1333153125               |
|                              | 61, 14, 4, 29, 51,                                 | —                        |
|                              | 37; (2)                                            | —                        |
|                              | 62, 13, 4, 4, (26)                                 | —                        |
|                              | 37; (1)                                            | —                        |
|                              | 4, (5, 14)                                         | 1324                     |
|                              | 63, 12, 23, 1, 8, 10, 1, 1, 2, 4, (9)              | 17 6456769682 0656739673 |
|                              |                                                    | 53237                    |
|                              | 37; 63, 9, 7, 49, 24, 17, (8)                      | —                        |
|                              |                                                    | 1523                     |
|                              |                                                    | 57673 29724              |
|                              |                                                    | 46653 69200              |
|                              | 64, 11, 5, 59, 8, 43, 31, 32, 41, 13, 16, (37, 37) | 17 66690                 |
|                              |                                                    | 2968377531               |
|                              | 37; 65, 10, 21, 2, 1, 1, 1, 2, 1, 4, 3, (12)       | 9                        |
|                              | 26, 25, 38, 31, 39, 22, 47, 15, 23, (6)            | 99046                    |
|                              | 37; 66, 1, 88, 2, 1, 33, (7)                       | 2 596189490              |
|                              | 7, 31, 21, —                                       | —                        |
| 1436                         | 37; 67, 9, 2, (18)                                 | —                        |
|                              | 35, (4)                                            | —                        |

TABLE 1001 TO 1500—CONTINUED.

| No. Continued-fraction lines |                                                                                    | $y / x$                          |
|------------------------------|------------------------------------------------------------------------------------|----------------------------------|
|                              | 68, —                                                                              | 256                              |
| 1437                         | 37; 1, 71, 1, 5, 2, 2, 4, 18, 1, 2, 1, 1, 1, (24)                                  | 42 83205 10399                   |
| 1438                         | 2, 29, 28, 17, 4, 53, 21, 41, 28, 47, (3)                                          | 64040                            |
| 1439                         | 69, 1, —                                                                           | —                                |
| 1440                         | 14, 1, 1, 7, 1, 3, 3, 24, 1, (36)                                                  | 91753956 903839 1849254806064    |
|                              | 37; 1, 9, 9, 1, 27, 47, 46, 9, 53, 18, 23, 3, 71, (2)                              | —                                |
|                              |                                                                                    | 07281 221757241139               |
|                              | 37; 2, 1, 5, 6, 1, 2, 1, 1, 2, 2, 1, 10, 7, 2, (37)                                | 338 95537 37610 64960            |
|                              | 70, (17) 5, 25, 46, 13, 11, 49, 22, 37, 35, 26, 23, 50, 7, 10, 35, (2)             | 12857 98517 00303                |
|                              | 37; 1, (4)                                                                         | —                                |
| 1441                         | 37; 9, 2, 1, 2, 2, 1, 3, 1, 3, 4, 1, 3, 1, 14, 2, 1, 1, 4, 2, (6)                  | 221 54985 95953 3851124040 54999 |
| 1442                         | 43, 8, 27, 40, 25, 24, 45, 17, 48, 19, 15, 51, 16, 57, 5, 29, 33, 40, 15, 32, (11) | 8410 14472 84306 8270308         |
| 1443                         | —                                                                                  | —                                |
|                              | 4, (36)                                                                            | 721                              |
| 1445*                        | 37; 1, —                                                                           | —                                |
| 1446                         | —                                                                                  | 1443                             |
| 1447                         | —                                                                                  | —                                |
| 1448                         | 73, (2)                                                                            | 1                                |
|                              |                                                                                    | 38                               |
|                              | 37; (74)                                                                           | 381                              |
|                              |                                                                                    | 38                               |
|                              | 38; (7, (61)) 2,                                                                   | —                                |

|                                                                            | No. | Continued-fraction lines                                                                                                                                                                                                                                                                                                                                                                                                                                               | $y \ / \ x$                                                                                                                                                                              |
|----------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                            |     | 38; (25, (38))<br>26, —<br><br>38; 7, 1, 3, 1, 1, 2, 3, 1, 5, 12, 1, 1, (37)<br>9, 54, 17, 39, 34, 27, 18, 51, 13, 6, 37, 39, (2)<br>38; (3, 4) 47,<br>1, 2, —                                                                                                                                                                                                                                                                                                         | 144159<br>3 42829 14389 10093<br>130 41033 17574<br>21848<br>—<br>—<br>—                                                                                                                 |
| 1449<br>1450*<br>1451<br>1452<br><br>1453*<br>1454<br>1455<br>1456<br>1457 |     | 38; 1, 6, 6, 1, 7, 2, 6, 4, (8)<br>12, 25, 25, 17, (9)<br>—<br>38; 10, 5, 1, 45, (8, 8)<br>38; 8, 4, 69, 1, 1, (9, 9)<br>10, 1, 1, 2, 2, 3, 1, 1, 2, 2, 14, 1, 4, 1, 1, (37)<br>—<br>—<br>—<br>8, 2, 1, 49, 23, 29, 19, 38, 35, 25, 31, 5, 59, 14, 35, 41, (2)                                                                                                                                                                                                         | 2773 0814284090<br>723<br>1 0811943933<br>—<br>31566 20707<br>85340 38 28530<br>—<br>—<br>—<br>32 50782 7844732                                                                          |
|                                                                            |     | 38; 7, (1, (84))                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2 626088778                                                                                                                                                                              |
|                                                                            |     | 38; 37, 39, 9, 1, 1, 5, 1, 3, 6, 10, 1, 2, 1, 2, 1, 1, 3, 18, 1, 3, 1,<br>1,<br>9, 31, 33, 41, 12, 51, 19, 12, 7, 52, 21, 44, 23, 36, 37, 21, 4,<br>57, 17, 36,<br>3, 1, 1, 2, 6, 2, 1, 2, (25, 25)<br>33, 28, 1, 27, 39, 28, (3, 3)<br>38; 10, 6, 1, 9, 2, 3, 1, 4, 1, 1, 2, 2, 4, 15, (38)<br>11, 67, 46, 31, 6, 37, 25, 29, 17, 5, (2)<br>38; 1, 25, (6)<br><br>38; 12, 11, (17, 1) 48, 9, (3)<br>38; 13, 5, 9, 55, (16)<br>8, 29, 32, 43, 7, 23, 16, (31)<br>56, — | —<br>54805<br>37 58165 73334 54563<br>1432 54652 50389 3444326 9776194930<br>96169 965981908898245<br>6707479<br>23766243<br>1<br>29855<br>18 54951<br>9341965238 3160730132<br>—<br>723 |

TABLE 1001 TO 1500—CONTINUED.

|                                                                                                                                                           | No. | Continued-fraction lines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $y \ / \ x$                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1458<br>1459<br>1460<br><br>1461<br>1462<br>1463<br>1464<br>1465*<br>1466<br>1467<br>1468<br>1469<br>1470<br>1471<br>1472<br>1473<br>1474<br>1475<br>1476 |     | 38; 5, 18, 1, 3, 1, 2, 1, 1, 7, 1, 10, (38)<br>—<br>—<br>38; 1, 49, 17, 26, 23, 34, 41, 9, 62, 7, (2)<br>13, 2, 1, 1, 7, 8<br>—<br>6, 39, 3, 37, 46, 1, 4, 2, 2, 2, 1, 1, 1, 5, 4, 15, 25, 2, 1, 1, (37)<br>45, 15, 29, 27, 25, 39, 30, 43, 13, 18, 5, 3, 30, 31, 45, (2)<br>—<br>—<br>—<br>18, 64, 2, 19, 16, 35, 4, 1, (1, (84))<br>—<br>—<br>18, 74, 49, 14, 1, 5, 3, 2, 1, 3, 3, 18, 1, 4, 6, 1, 2, 1, (24)<br>41, —<br>—<br>3, 61, 12, 20, 25, 44, 19, 23, 4, 59, 15, 11, 51, 20, 55, (3)<br>—<br>1884, 33, 5, 31, — | 88890 25540<br>27540<br>22872<br>25694 24820 3012 96401<br>78 52281<br>—<br>81439 56337 9998412034 45032 81010<br>9 239 25898<br>—<br>—<br>—<br>6<br>15529<br>10897<br>33506<br>4 16529 16197 65845 0827252785<br>—<br>97289<br>—<br>2753 |

| No.  |                                                                     | Continued-fraction lines | $y / x$                            |
|------|---------------------------------------------------------------------|--------------------------|------------------------------------|
| 1477 | (-74)                                                               |                          | —                                  |
| 1478 | —                                                                   |                          | —                                  |
|      | 1894, 1, —                                                          |                          | 4                                  |
|      | 2308, 1, 15, 2, 1,                                                  |                          | 23712593                           |
|      | —                                                                   |                          | 72                                 |
|      | 25, 47, (1, (25))                                                   |                          | 9 07925                            |
|      | —                                                                   |                          | 14427                              |
|      | 1383, 49, 2, 4, 1, 2, 1, 1, 4, 4, 1, 7, 1, 2, (3, 3)                |                          | 2 0726490 21285                    |
|      |                                                                     |                          | 82 84932                           |
|      | 39, 16, 15, 56, 9, 49, 24, (21, 21)                                 |                          | —                                  |
|      | 1382, 40, 31, 6, 39, 24, 17, 4, 26, 2, 35, 2, (2, 2)                |                          | 1533 05185                         |
|      | 11, 17, (25, 25)                                                    |                          | 16                                 |
|      |                                                                     |                          | 52043                              |
|      | 1383, 1, 23, 1, (3, (72))                                           |                          | 73045                              |
|      | 46, 27, 41,                                                         |                          | 80026                              |
|      |                                                                     |                          | 640                                |
|      | 2843, 1, 13, 1, 1, 3, 49, 25, 9, 1, 1, 5, 1, 6, 8, 2, 1, 2, 1, 1,   |                          | 18883 73036 35216                  |
|      | (18)                                                                |                          |                                    |
|      | 18, 36, 19, 3, 8, 39, 37, 12, 57, 11, 9, 27, 41, 24, 33, 43, (4)    |                          | 46281 85247                        |
|      |                                                                     |                          | 64922                              |
|      |                                                                     |                          | 7 23520 34792                      |
|      | (1, (34)) 853, 5, 369, 33,                                          |                          | 19836                              |
|      |                                                                     |                          | 7 60216358                         |
|      | 2682, (11) 4, (5)                                                   |                          | —                                  |
|      | 49, —                                                               |                          | 5291                               |
|      | 328, 1, 4, 1, 4, 3, 2, 4, 12, 1, 1, 3, 1, 2, 1, 6, 1, 14, 2, 8, 25, |                          | 36737                              |
|      | 17, 6, 41, 35, 18, 47, 21, 51, 10, 63, 5, 35, 9, 3,                 |                          | 66160                              |
|      | 27, 45, 1, 14, 53, 15, 21, 30, 1, 1, 10, 2, 1, (37)                 |                          | 89163 97276 23871 56715            |
|      | 30, 45, 7, 25, 51, (2)                                              |                          | 1 01712 37347 28288 54875 60183 30 |
|      | 28, 1, 34, 1, 15, 42, 2, 4, 1, 1, 3, 17, 1,                         |                          | 39 01057                           |
|      | 38; 2, (2)                                                          |                          | 1151                               |
|      | 2922, 41, (2, 3) 1, 2, 2, 2, 9, 5, 1, 3, 1, (24)                    |                          | 2656 14220                         |
|      | 24, 31, 8, 13, 53, 16, 59, (3)                                      |                          | —                                  |
|      | 39, 23, 43, 144, 57, 7, 1, 2, 2, 4, 1, 2, 3, 1,                     |                          | 8962979326 93399                   |
|      | 38; 2, 37, 6, 9, 1, 50, 23, 30, 15, 46, 25, 18, 55, 10, 5, (38)     |                          | 1 43151 08109                      |
|      | 3082, 1, 1, 7, 15, (2)                                              |                          | 95957 60507 79476                  |
|      | 2, 32, (2, (52))                                                    |                          | 29745                              |
|      | 2393, 41, 11, —                                                     |                          | 54 6114769298                      |
|      | 41, 34, —                                                           |                          | 5                                  |
|      |                                                                     |                          | 83                                 |
|      | 3812, 25, 6, 1, 2,                                                  |                          | 2 18560                            |
|      | 20, (9)                                                             |                          | —                                  |
|      | 37, 23, 3, (8)                                                      |                          | —                                  |
|      | 3126, 8, 2,                                                         |                          | —                                  |
|      | 3282, 21, 415, 1, 1, 1, 1, 5, 1, 3, 1, 2, 18, 1, 6, 25, 2, (10)     |                          | 2 61633 47062 2926284001           |
|      | 9, 28, 37, 33, 36, 31, 43, 12, 53, 17, 44, 27, 4, 63, 11, 3, 36,    |                          | 100 55043 41504 36351              |
|      | (7)                                                                 |                          |                                    |
|      | 17, 3,                                                              |                          | 728102117                          |
|      | 3832, (38)                                                          |                          | 1 193711174                        |
|      | 34, 4, (2)                                                          |                          | 59                                 |

TABLE 1001 TO 1500—CONTINUED.

| No.   |                                  | Continued-fraction lines | $y / x$  |
|-------|----------------------------------|--------------------------|----------|
| 1479  | 38; 2, 5, 2, 2, 1, 1, 1, (1)     |                          | 1 14525  |
| 1480  | 35, 13, 30, 25, 38, 33, 35, (34) |                          | 44 04376 |
| 1481* | —                                |                          | 948 5559 |

|       | No. | Continued-fraction lines                                                                    | $y \ / \ x$                  |
|-------|-----|---------------------------------------------------------------------------------------------|------------------------------|
| 1482  | 38; | 2, 8, (19)                                                                                  | —                            |
| 1483  | 36, | 9, (4)                                                                                      | 2 13859                      |
| 1484  | —   |                                                                                             | —                            |
| 1485  | 38; | 2, 14, 1, 8, 1, 2, 5, 1, 1, 2, 1, 4, 10, 1, 3, 1, 1, (1, 1)                                 | 1 01945 97513 61405          |
| 1486  | 37, | 5, 65, 8, 49, 25, 13, 40, 35, 23, 47, 16, 7, 56, 17, 41, 32, (35, 35)                       | 39 23264 72655 17468         |
| 1487  | —   |                                                                                             | —                            |
|       | 38; | (2) (38)                                                                                    | 2                            |
|       |     |                                                                                             | 77                           |
|       | 38; | 1, 1, 25, 5, 1, 7, 1, 2, 1, 1, 1, 1, 2, 4, 6, 1, 3, 2, 2, 1, 1, 12, 3, 1, (37)              | 41 20928 70570 44632 38045   |
|       | 39, | 38, 3, 13, 58, 9, 51, 22, 39, 33, 34, 37, 27, 17, 1, 53, 18, 29, 26, 33, 43, 6, 19, 57, (2) | 1586 95889 00942 30191 81226 |
|       | 38; | 1, 1, (10)                                                                                  | 44                           |
|       | 40, | 37, (?)                                                                                     | 1695                         |
|       |     |                                                                                             | 1079                         |
|       |     |                                                                                             | 28                           |
|       | 38; | 1, 1, (6)                                                                                   | —                            |
|       | 41, | 36, (1)                                                                                     | —                            |
|       | 38; | 1, 1, 4, 1, 1, 1, 3, (38)                                                                   | 4 92228                      |
|       | 42, | 35, 15, 43, 30, 41, 21, (2)                                                                 | 189 74735                    |
|       |     |                                                                                             | 2 05495                      |
|       |     |                                                                                             | 79 24232145                  |
|       | 38; | 1, 1, 3, 1, 1, 3, 1, (37)                                                                   | —                            |
|       | 43, | 34, 19, 37, 38, 17, 59, (2)                                                                 | —                            |
| 1488  | 44, | 33, 23, 49, (1)                                                                             | 12151                        |
|       |     |                                                                                             | 25 78145                     |
|       |     |                                                                                             | 994 84332                    |
| 1489* | 38; | 1, 1, 2, 2, 1, 4, 2, 3, 1, (1, 1)                                                           | 5                            |
| 1490* | 45, | 32, 27, 24, 47, 15, 31, 19, 40, (33, 33)                                                    | 1716973                      |
| 1491  | —   |                                                                                             | —                            |
| 1491b | 38; | 1, (1, 1)                                                                                   | 68230                        |
|       | 46, | (31, 30)                                                                                    | 13 56270                     |
|       |     |                                                                                             | 523 87849                    |
|       | 38; | 1, 1, 1, 1, 2, 2, (1)                                                                       | —                            |
|       | 47, | 30, 35, 37, 26, 25, (42)                                                                    | —                            |
| 1492  | 38; | 1, 1, 1, 2, 10, 1, 1, 1, (18)                                                               | —                            |
|       | 48, | 29, 39, 28, 7, 48, 27, 49, (4)                                                              | —                            |
| 1493* | 38; | 1, 1, 1, 3, 2, 2, 18, 1, (10, 10)                                                           | 1694 41225                   |
| 1494  | 49, | 28, 43, 19, 28, 31, 4, 67, (7, 7)                                                           | 65471 00182                  |
|       | 38; | 1, 1, 1, 7, 15, 5, (38)                                                                     | 17040490 0052251645          |
|       | 50, | 27, 47, 10, 5, 25, (2)                                                                      | 118673                       |
| 1495  | 38; | 1, (1)                                                                                      | 3365                         |
|       | 51, | (26)                                                                                        | —                            |
| 1496  | 38; | 1, 2, (9)                                                                                   | —                            |
|       | 52, | 25, (8)                                                                                     | —                            |
| 1497  | 38; | 1, 2, 4, 4, 1, 1, 1, 1, 6, 2, 2, 1, 8, 1, (24)                                              | 58 19614 30932               |
| 1498  | 53, | 24, 17, 16, 41, 33, 32, 43, 11, 31, 23, 51, 8, 67, (3)                                      | 2251 67187 51127             |
| 1499  | 38; | 1, 2, 2, 1, 1, 1, 3, 1, 12, 8, 1, 1, (10)                                                   | 83118 18384                  |
| 1500  | 54, | 23, 26, 39, 31, 42, 17, 57, 6, 9, 38, 39, (?)                                               | 32 17006 59967               |
|       | 38; | 1, 2, 1, 1, 7, 5, 1, 4, 1, 2, 3, 1, 2, 1, 1, 2, 10, 1, 2, 15, 6, 1, (37)                    | 77552 05839 43067 80299      |
|       | 55, | 22, 35, 41, 10, 13, 55, 14, 47, 25, 19, 46, 23, 38, 35, 29, 7, 49, 26, 5, 11, 65, (2)       | 30 02576 94656 26953 60610   |
|       | 38; | 1, 2, 1, 2, 2, 1, 6, 2, 1, (18)                                                             | 118 11844                    |
|       | 56, | 21, 44, 25, 24, 49, 11, 25, 51, (4)                                                         | 4574 70751                   |

In connexion with the subject we have a paper, “A Table of the Square Roots of Prime Numbers of the form

$4m + 1$  less than 10000 expanded as Periodic Continued Fractions,” by C. A. Roberts, with Introduction and Explanation by Artemas Martin, the *Mathematical Magazine*, vol. II. (No. 7, for October, 1892), pp. 105–120. This extends, in fact, to numbers up to 10501, but only the denominators of the continued fractions (that is, the first lines of Degen’s and the present table) are given: thus the entry for 1009 is 31; 1, (3, 3).

The paper just referred to notices errors in Degen’s tables for the numbers 853 and 929. For 853 the first line should be

$$29, 4, 1, 5, 1, 2, 4, 1, 1, 15, 19, (2, 2),$$

(15 instead of Degen’s 14). For 929 the first and second lines should be

$$30, 2, 11, 1, 2, 3, 2, 7, 5, (2, 2)$$

$$1, 29, 5, 40, 19, 16, 25, 8, 11, (23, 23).$$

The values of  $x, y$  in Table I. and those in Table II. (for the solution of  $y^2 = ax^2 - 1$ ) are correct for each of the numbers 853 and 929.

# ON HALPHEN'S CHARACTERISTIC $n$ , IN THE THEORY OF CURVES IN SPACE.

[949]

[From *Crelle's Journal d. Mathematik*, t. CXI. (1893), pp. 347–352.]

If we consider a curve in space without actual singularities, of the order (or degree)  $d$ , then this has a number  $h$  of apparent double points (adps.), viz. taking as vertex an arbitrary point in space, we have through the curve a cone of the order  $d$ , with  $h$  nodal lines; and Halphen denotes by  $n$  the order of the cone of lowest order which passes through these  $h$  lines. For a given value of  $d$ ,  $h$  is at most  $= \frac{1}{2}(d-1)(d-2)$ , and as shown by Halphen it is at least  $= [\frac{1}{4}(d-1)^2]$ , if we denote in this manner the integer part of  $\frac{1}{4}(d-1)^2$ . For given values of  $d$ ,  $h$ , it is easy to see that  $n$  must lie within certain limits, viz. if  $\nu$  be the smallest number such that  $\frac{1}{2}\nu(\nu+3)$  equal to or greater than  $h$ , then  $n$  is at most  $= \nu$ ; and moreover  $n$  must have a value such that  $nd$  is at least  $= 2h$ , or say we must have  $nd = 2h + \theta$ , where  $\theta$  is  $= 0$  or positive. For any given value of  $d$ , we thus have a finite number of forms  $(d, h, n)$ , and we have thus *primâ facie* curves in space of the several forms  $(d, h, n)$ : but it may very well be, and in fact Halphen finds, that when  $d = 9$  or upwards, then for certain values of  $h$ ,  $n$  as above, there is not any curve  $(d, h, n)$ ; thus  $d = 9$ ,  $h = 17$  the values of  $n$  are  $n = 4$ ,  $n = 5$ , but there is not any curve  $d = 9$ ,  $h = 17$  for either of these values of  $n$ ; or say the curves  $(9, 17, 4)$  and  $(9, 17, 5)$  are non-existent. And in the Notes and References to the papers 302, 305 in vol. v. of my *Collected Mathematical Papers*, 4to. Cambridge, 1892, see p. 615, I remarked that, in certain cases for which Halphen finds a curve  $(d, h, n)$ , such curve does not exist except for special configurations of the  $h$  nodal lines not determined by the mere definition of  $n$  as the order of the cone of lowest order which passes through the  $h$  nodal lines; for instance  $d = 9$ ,  $h = 16$ ,  $n = 4$ , for which Halphen gives a curve, I find that for the existence of the curve it is not enough that the 16 lines are situate upon a quartic cone, but they must be the 16 lines of intersection of two quartic cones.

In fact, starting from an existing curve, say the complete intersection of two given surfaces of the orders  $\mu$ ,  $\nu$  respectively, we have, as is known,

$$d = \mu\nu, \quad 2h = \mu\nu(\mu-1)(\nu-1) :$$

we find also, as will appear,  $n = (\mu-1)(\nu-1)$ : hence also  $nd = 2h$ , viz. the cone of the order  $n$  through the  $h$  nodal lines meets the cone of the order  $d$  in these lines counting each twice, and in no other lines. I remark also that  $h$  is  $= \frac{1}{2}n(n+3)$  if  $\mu + \nu = 4$ ; viz. we have here two nodal lines lying in a plane; but if  $\mu + \nu > 4$ , then  $h$  is greater than  $\frac{1}{2}n(n+3)$ , viz. in this case the nodal lines are not  $h$  lines taken at pleasure, but they are lines subject to the condition of lying in a cone of the order  $n$ . But this is not all; the  $h$  nodal lines lie not only in this cone of lowest order  $n$ , but also in cones of the orders  $n+1$ ,  $n+2$ ,  $\dots$ ,  $n + (\mu + \nu - 2)$  respectively: I do not for the moment attempt to determine the number of independent conditions which are hereby imposed upon the  $h$  lines.

The last-mentioned theorem constitutes in fact the geometrical interpretation of results contained in Jacobi's paper "De eliminatione variabilis e duabus aequationibus algebraicis," *Crelle*, t. xv. (1836), pp. 101–124, [*Ges. Werke*, t. III. pp. 295–320]. Consider the two equations

$$U = (*|x, y, z, w)^\mu = 0; \quad V = (*|x, y, z, w)^\nu = 0;$$

representing surfaces of the orders  $\mu$ ,  $\nu$  respectively; since the form of the equations is quite arbitrary, we may without loss of generality assume that the point

$$(x, y, z) = (0, 0, 0)$$

is an arbitrary point in space; and this being so, we find the equation of the cone, vertex this point, which passes through the curve of intersection of the two surfaces by the mere elimination of  $w$  between the two equations. As the reasoning is exactly the same for a particular case, I write for convenience  $\mu = 3$ ,  $\nu = 4$ , and consider the two equations

$$A_0w^3 + A_1w^2 + A_2w + A_3 = 0, \quad B_0w^4 + B_1w^3 + B_2w^2 + B_3w + B_4 = 0,$$

where the suffixes show the degrees in regard to  $(x, y, z)$ , viz.  $A_0$ ,  $B_0$  are mere constants,  $A_1$ ,  $B_1$  are linear functions  $(*|x, y, z)^1$ ,  $A_2$ ,  $B_2$  quadric functions  $(*|x, y, z)^2$ , and so on. Multiplying the first equation

successively by  $1, w, w^2, w^3$ , and the second equation successively by  $1, w, w^2$ , we have 7 equations from which to eliminate  $1, w, w^2, w^3, w^4, w^5, w^6$ , and the result is

$$\begin{vmatrix} A_0 & A_1 & A_2 & A_3 & \cdot & \cdot & \cdot \\ \cdot & A_0 & A_1 & A_2 & A_3 & \cdot & \cdot \\ \cdot & \cdot & A_0 & A_1 & A_2 & A_3 & \cdot \\ \cdot & \cdot & \cdot & A_0 & A_1 & A_2 & A_3 \\ B_0 & B_1 & B_2 & B_3 & B_4 & \cdot & \cdot \\ \cdot & B_0 & B_1 & B_2 & B_3 & B_4 & \cdot \\ \cdot & \cdot & B_0 & B_1 & B_2 & B_3 & B_4 \end{vmatrix} = 0,$$

viz. this is an equation

$$(*|x, y, z)^{12} = 0,$$

of the cone of the order  $d = 12$ , through the curve of intersection of the two surfaces: say this equation is  $\Omega = 0$ .

But if we only multiply the first equation by  $1, w, w^2$  successively and the second equation by  $1, w$  successively, then, we have 5 equations serving to determine the ratios of  $w^5, w^4, w^3, w^2, w, 1$ , viz. we have these quantities proportional to the six determinants which can be formed out of the matrix

$$\begin{vmatrix} A_0 & A_1 & A_2 & A_3 & \cdot & \cdot \\ \cdot & A_0 & A_1 & A_2 & A_3 & \cdot \\ \cdot & \cdot & A_0 & A_1 & A_2 & A_3 \\ B_0 & B_1 & B_2 & B_3 & B_4 & \cdot \\ \cdot & B_0 & B_1 & B_2 & B_3 & B_4 \end{vmatrix},$$

say we have

$$w^5 : w^4 : w^3 : w^2 : w : 1 = L : M : N : P : Q : R,$$

where  $L, M, N, P, Q, R$  represent homogeneous functions  $(*|x, y, z)^e$ , of the degrees 11, 10, 9, 8, 7, 6 respectively. We may if we please write

$$\frac{w^5}{L} = \frac{w^4}{M} = \frac{w^3}{N} = \frac{w^2}{P} = \frac{w}{Q} = \frac{1}{R},$$

or eliminating  $w$ , we have the series of equations which may be written

$$\begin{vmatrix} L & M & N & P & Q \\ M & N & P & Q & R \end{vmatrix} = 0,$$

viz. we thus denote that the determinants formed with any two columns of this matrix are severally  $= 0$ . This of course implies that each of the determinants in question is the product of  $\Omega$  and a factor which

is a homogeneous function of the proper degree in  $(x, y, z)$ , so that the several equations are all of them satisfied if only  $\Omega = 0$ . We have for instance  $PR - Q^2 = A\Omega$ , where  $A$  is a quadric function  $(*|x, y, z)^2$ ; similarly  $NR - PQ = B\Omega$ , where  $B$  is a cubic function  $(*|x, y, z)^3$ ; and the like as regards the other determinants.

If the ratios  $x : y : z$  have any given values such that we have for these  $\Omega = 0$ , then  $w$  has a determinate value, that is, on each line of the cone  $\Omega = 0$ , there is a single point of the curve of intersection of the two surfaces: the only exceptions are when, for the given values of  $x : y : z$ , the expressions for  $w$  assume an indeterminate form, viz.  $w$  has then two values, and there are upon the line two points of the curve, or what is the same thing, the line is a nodal line of the cone: the conditions for a nodal line thus are  $L = 0$ ,  $M = 0$ ,  $N = 0$ ,  $P = 0$ ,  $Q = 0$ ,  $R = 0$ , viz. each of these equations is that of a cone passing through the nodal lines of the cone  $\Omega = 0$ ; the cone of lowest order is  $R = 0$ , a cone of the order 6 meeting the

cone  $\Omega = 0$  of the order 12 in 36 lines each twice, which lines are consequently the nodal lines of the cone  $\Omega = 0$ . The mere condition of the 36 lines lying upon a cone of the order 6 shows that the 36 lines are not arbitrary; and we have moreover, through the 36 lines, cones of the orders 7, 8, 9, 10 and 11 respectively. Obviously the foregoing reasoning is quite general, and for the surfaces of the orders  $\mu, \nu$  we have, as stated above, the cone  $\Omega$  of the order  $\mu\nu$ , with  $h = \frac{1}{2}\mu\nu(\mu-1)(\nu-1)$  nodal lines, the intersections (each counting twice) of this cone with a cone of the order  $n = (\mu-1)(\nu-1)$ ; and moreover the  $h$  nodal lines lie also in cones of the orders  $n+1, n+2, \dots, n+\mu+\nu-2$  respectively.

To examine the meaning of the theorem, I form the table

| $\mu, \nu$ | $d$ | $h$ | $n, \dots, n + \mu + \nu - 2$  |
|------------|-----|-----|--------------------------------|
| 2, 2       | 4   | 2   | 1, 2, 3                        |
| 2, 3       | 6   | 6   | 2, 3, 4, 5                     |
| 2, 4       | 8   | 12  | 3, 4, 5, 6, 7                  |
| 2, 5       | 10  | 20  | 4, 5, 6, 7, 8, 9               |
| $\vdots$   |     |     |                                |
| 3, 3       | 9   | 18  | 4, 5, 6, 7, 8                  |
| 3, 4       | 12  | 36  | 6, 7, 8, 9, 10, 11             |
| 3, 5       | 15  | 60  | 8, 9, 10, 11, 12, 13, 14       |
| $\vdots$   |     |     |                                |
| 4, 4       | 16  | 72  | 9, 10, 11, 12, 13, 14, 15      |
| 4, 5       | 20  | 120 | 12, 13, 14, 15, 16, 17, 18, 19 |

Here  $\mu, \nu = 2, 2$ , there are 2 nodal lines, which are arbitrary, and of course lie on cones of the orders 1, 2, 3 respectively. So  $\mu, \nu = 2, 3$ ; there are 6 nodal lines, which are not arbitrary, inasmuch as they lie on a cone of the order 2; but regarding them as arbitrary lines on such a cone, we can through them draw a cone of the order 3 or any higher order, and it is thus no specialisation to say that they lie upon cones of the orders 3, 4, and 5. But going a step further  $\mu = 2, \nu = 4$ : here we have 12 nodal lines which, inasmuch as they lie on a cone of the order 3, are not arbitrary: and they are not arbitrary lines upon this cone, for they lie on a cone of the order 4, and such a cone can be drawn through at most 11 arbitrary lines on a cubic cone. In fact, upon a cone of the order  $\theta$ , taking at pleasure  $N$  lines, the condition that it may be possible through these to draw a cone of the order  $\theta + 1$  is  $\frac{1}{2}(\theta + 1)(\theta + 4) = N + 3$  at least; for if this number were  $= N + 2$ , then through the  $N$  points we have only the improper cone  $(\alpha x + \beta y + \gamma z) U_\theta = 0$ , if  $U_\theta = 0$  is the cone of the order  $\theta$ . It thus appears that the 12 nodal lines are not arbitrary lines on a cubic cone, but that they constitute the complete intersection of a cubic cone and

a quartic cone. But through such 12 lines we may draw cones of the 5th and higher orders, and it is thus no further condition that the 12 lines lie on cones of the orders 5, 6 and 7 respectively.

So again  $\mu = 3, \nu = 3$ ; we have here 18 nodal lines which, inasmuch as they lie on a cone of the order 4, are not arbitrary: and they are not arbitrary lines on this cone inasmuch as they lie also on a cone of the order 5, and such a cone can be drawn through at most 17 arbitrary lines on the quartic cone: it thus appears that the 18 nodal lines are 18 out of the 20 lines of intersection of a quartic cone and a quintic cone. But there is no further condition, for through such lines we can draw a cone of the order 6 or any higher order

and thus the lines lie on cones of the orders 6, 7 and 8 respectively. It appears probable however that, for higher values of  $\mu, \nu$ , it would be necessary to take account not only (as in these examples) of the cones of the orders  $n$  and  $n + 1$ , but of those of higher orders  $n + 2$ , &c.; and thus that it is not the true form of the theorem to say that the  $h$  nodal lines must be  $h$  out of the  $n(n + 1)$  lines of intersection of two cones of the orders  $n$  and  $n + 1$  respectively.

It appears, by what precedes, that the  $h, = \frac{1}{2}\mu\nu(\mu - 1)(\nu - 1)$ , lines which are the nodal lines of the cone of arbitrary vertex which passes through the curve of intersection of two surfaces of the orders  $\mu, \nu$  respectively, form a remarkable special system of lines, which well deserve further study. I remark also that, without having proved the negative, it seems to me clear that given the values of  $d, h, n$  it is only in the cases where the  $h$  lines form some such special system (and not in the general case where the  $h$  nodal lines are any lines whatever on a cone of the order  $n$ ) that there exists a curve  $(d, h, n)$ ; and thus that the question for further investigation is, for given values of  $(d, h, n)$  to determine the conditions necessary for the existence of a curve in space with these characteristics  $(d, h, n)$ .

# ON THE SEXTIC RESOLVENT EQUATIONS OF JACOBI AND KRONECKER.

[950]

[From *Crelle's Journal d. Mathematik*, t. CXIII. (1894), pp. 42–49.]

The equations referred to are: the first of them, that given by Jacobi in the paper “Observatiunculæ ad theoriam æquationum pertinentes,” *Crelle*, t. XIII. (1835), pp. 340–352, [*Ges. Werke*, t. III., pp. 269–284], under the heading “Observatio de æquatione sexti gradus ad quam æquationes sexti gradus revocari possunt,” and the second, that of Kronecker in the note “Sur la résolution de l'équation du cinquième degré,” *Comptes Rendus*, t. XLVI. (1858), pp. 1150–1152. Jacobi's equation is closely connected with that obtained by Malfatti in 1771, see Brioschi's paper “Sulla risolvente di Malfatti per l'equazione del quinto grado,” *Mem. R. Ist. Lomb.*, t. IX. (1863); but the characteristic property first presents itself in Jacobi's form, and I think the equation is properly described as Jacobi's resolvent equation. The other equation has been always known as Kronecker's resolvent equation; it belongs to the class of equations for the multiplier of an elliptic function considered by Jacobi in the paper “Suite des notices sur les fonctions elliptiques,” *Crelle*, t. III. (1828), pp. 303–310, see p. 308, [*Ges. Werke*, t. I., pp. 255–263, see p. 261]: say Kronecker's equation belongs to the class of Jacobi's Multiplier Equations. We have in regard to it the paper by Brioschi, “Sul metodo di Kronecker per la risoluzione delle equazioni di quinto grado,” *Atti Ist. Lomb.*, t. I. (1858), pp. 275–282, and see also the “Appendice terza” to his translation of my *Elliptic Functions* (Milan, 1880): it seems to me however that the theory of Kronecker's equation has not hitherto been exhibited in the clearest form.

I consider the forms

|       |       |
|-------|-------|
| 12345 | 13524 |
| 13254 | 12435 |
| 24315 | 23541 |
| 35421 | 34152 |
| 41532 | 45213 |
| 52143 | 51324 |

which, if in the first instance the figures are regarded as points, represent the twelve pentagons which can be formed with the points 1, 2, 3, 4, 5; each form in the right-hand column is derived from the corresponding form in the left-hand column by stellation, say we have  $13524 = S\,12345$ , and so in other cases.

A pentagon is in general reversible, but we sometimes consider it as irreversible (viz. we distinguish between the pentagons 12345 and 15432); when this is so, we write  $15432 = R\,12345$ , and we have thus twelve new forms, in all twenty-four forms. The symbols  $R$ ,  $S$  are such that  $R^2 = 1$ ,  $S^2 = R$ ,  $RS = SR$ ,  $S^4 = 1$ . But for a reversible pentagon, there is no occasion to use the symbol  $R$ , and we have simply  $S^2 = 1$ .

In a somewhat different point of view, we may for an irreversible pentagon write

|            |                                |                                    |
|------------|--------------------------------|------------------------------------|
| 12345      | to denote any one of the forms | 12345, 23451, 34512, 45123, 51234, |
| $R\,12345$ | to denote any one of the forms | 15432, 54321, 43215, 32154, 21543, |

and for a reversible pentagon 12345 to denote any one of these same ten forms.

Each pentagon gives thus ten forms, viz. we have in all 120 forms which all are the different arrangements of the five figures. But further regarding the arrangement 12345 as positive, then the forms in the left-hand column are each of them positive, and the ten forms derived from any one of these are each of them positive, that is, the forms in the left-hand column give all the 60 positive arrangements of the five figures: and similarly the forms in the right-hand column give all the 60 negative arrangements of the five figures.

Taking 1, 2, 3, 4, 5 to denote any quantities, or say the five roots  $x_1, x_2, x_3, x_4, x_5$  of a quintic equation, we regard 12345, . . . , as denoting functions of these roots: in particular, 12345 may denote a cyclic reversible function, the analogue of the reversible pentagon, or it may denote a cyclic irreversible function, the analogue of the irreversible pentagon.

*Jacobi's resolvent equation.*

The most simple course is to take 12345 a cyclic reversible function of the roots  $x$ ; a root of the resolvent equation is then  $12345 - 13524, = (1 - S) 12345$ , and the six roots are

$$z_1 = (1 - S) 12345,$$

$$z_2 = (1 - S) 13254,$$

$$z_3 = (1 - S) 24315,$$

$$z_4 = (1 - S) 35421,$$

$$z_5 = (1 - S) 41532,$$

$$z_6 = (1 - S) 52143.$$

Here effecting on the roots  $x$  any positive substitution whatever, we permute *inter se* the roots  $z$ ; but effecting on the roots  $x$  any negative substitution whatever, then [the discussion continues on the following page, outside the present block].